Dynamic Vacuum Coupling Hypothesis (DVCH): A UV-Regularized Interacting-Vacuum Framework with Background Closure, Global Stability, and Diagnostic Tests

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(Dated: August 29, 2026)

We present a formulation of the *Dynamic Vacuum Coupling Hypothesis* (DVCH), a nonlinear interacting-vacuum scenario treated as an effective macroscopic closure within a covariant interacting dark-sector framework. The model combines power-law tracking between vacuum and matter densities with an ultraviolet (UV) curvature suppression inspired by higher-curvature effective field theory (EFT). We (i) provide a consistent covariant action-based motivation for dark-sector interaction via conformal matter coupling to a scalar EFT, (ii) derive the exact implicit background system in terms of $E(z) = H(z)/H_0$ and obtain the exact energy-transfer function Q implied by the DVCH closure, (iii) show that the effective vacuum equation of state satisfies $w_{\Lambda,\mathrm{eff}} > -1$ whenever $Q < 0$, while any numerical phantom-divide crossing must correspond to a sign change of Q or to an inconsistent plotting convention, (iv) derive the conditions under which the autonomous background system approaches a Minkowski end state, and (v) formulate linear perturbation equations, low-redshift growth diagnostics, a parameter-space robustness scan, and an exploratory public-data likelihood diagnostic implementation for Pantheon+, DESI BAO, and cosmic-chronometer measurements. The CMB discussion is restricted to literature benchmarks and to the source-table interface used to compare the background sector against Planck 2018 literature benchmarks. In this form, DVCH is presented as an effective interacting-vacuum framework whose observational status remains to be established by full likelihood analyses.

I. INTRODUCTION

The standard ΛCDM model, while remarkably successful, is under intense scrutiny because of the persistent "Hubble tension"—a 5σ discrepancy between early-universe measurements from Planck and local late-time observations from SH0ES [1, 2]. In addition, the S_8 tension suggests that the Universe is less "clumpy" than predicted by the baseline model [3, 4].

Interacting Dark Energy (IDE) models, including the Dynamic Vacuum Coupling Hypothesis (DVCH) developed here, provide a useful laboratory for testing whether nontrivial energy-momentum exchange Q between dark sectors can remain both theoretically consistent and observationally viable [5–7]. Unlike ad hoc parameterizations that can trigger early-time instabilities, DVCH adopts a macroscopic closure motivated by ultraviolet (UV) curvature regularization and leads to a qualitatively distinct late-time Minkowski attractor. Here the goal is to define an internally consistent interacting-vacuum closure and to document background, stability, growth, and exploratory likelihood diagnostics [8–10].

This work develops DVCH as a *macroscopic closure* for a covariant interacting dark sector. The guiding philosophy is explicit: we do not claim uniqueness of a UV-complete microphysical realization, but we do require internal consistency, correct conservation-law bookkeeping, stable perturbative prescriptions, and clear numerical implementability.

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Validation framework for DVCH

This work records a set of consistency checks for DVCH models, combining the covariant action-based motivation, the macroscopic background closure, the exact interaction kernel, the autonomous-system stability analysis, and the structure-growth diagnostic. The components treated here are the covariant motivation, the background closure, the exact interaction kernel, the global stability analysis, the sub-horizon growth calculation, the (n, β) background robustness scan, and an exploratory late-time public-data fit.

The CMB/Planck requirements are represented explicitly in the diagnostic package: a DVCH background/source-table interface for CLASS/CAMB-style backend checks [11, 12], the photon–baryon/CMB component checklist, angular-spectrum and lensing targets, a Planck `clik` likelihood configuration target [13], and a Cobaya-compatible Planck-stack configuration target [14]. Consequently, every observational number below is a diagnostic quantity unless explicitly stated otherwise. The aim is to state the assumptions and limitations of the UV-regularized interacting-vacuum closure and to separate local calculations from CMB literature benchmarks.

II. COVARIANT THEORETICAL BASIS: AN INTERACTING DARK SECTOR FROM AN EFT-MOTIVATED ACTION

A frequent referee objection to IDE phenomenology is the absence of a covariant field-theory structure explaining how an interaction Q arises. A minimal and standard approach is to embed the interaction in a scalar-tensor EFT where cold matter couples conformally to the scalar while radiation remains minimally coupled. This produces an *unambiguous* covariant energy-exchange vector Q^μ.

A. EFT-inspired action (covariant, minimal)

We consider (in units $\hbar = c = 1$ unless explicitly restored)

$$S = \int \mathrm{d}^4x\,\sqrt{-g}\left[\frac{M_{\rm Pl}^2}{2}R - \frac{1}{2}(\nabla\phi)^2 - V(\phi) + \frac{\gamma}{2}\ell_{\rm P}^2 R^2\right] + S_m\left[\psi_m,\, A^2(\phi)\, g_{\mu\nu}\right] + S_r[\psi_r,\, g_{\mu\nu}]\,.$$

Here $A(\phi)$ is a conformal coupling function. The R^2 term is the simplest local higher-curvature EFT correction, in the same broad class of curvature corrections used in early-universe model building [15]; its role is *motivational* for a high-curvature suppression scale and does not dominate at observable redshifts.

Varying the action yields

$$\nabla_\mu T^{\mu\nu}_{(m)} = +Q^\nu, \qquad \nabla_\mu T^{\mu\nu}_{(\phi)} = -Q^\nu, \qquad \nabla_\mu T^{\mu\nu}_{(r)} = 0, \tag{1}$$

with the standard scalar-tensor exchange vector

$$Q^\nu = \left(\frac{\mathrm{d}\ln A}{\mathrm{d}\phi}\right) T_{(m)}\,\nabla^\nu\phi, \qquad T_{(m)} \equiv T^\mu{}_{(m)\mu} \simeq -\rho_m. \tag{2}$$

In FLRW, $Q^\nu = Qu^\nu$ with

$$Q = -\left(\frac{\mathrm{d}\ln A}{\mathrm{d}\phi}\right)\rho_m\dot{\phi}. \tag{3}$$

Equation (3) provides a *covariant* origin for energy transfer. DVCH will now be formulated as a *closure relation* for ρ_Λ (interpreted as the effective vacuum-like dark energy), which in turn determines Q at the background level.

B. Macroscopic closure vs. microscopic model

At the level of homogeneous cosmology, many microphysical models map into an effective two-fluid system:

$$\dot{\rho}_r + 4H\rho_r = 0, \tag{4}$$
$$\dot{\rho}_m + 3H\rho_m = -Q, \tag{5}$$
$$\dot{\rho}_\Lambda = Q, \tag{6}$$

where ρ_Λ is vacuum-like in the sense of a baseline pressure $p_\Lambda = -\rho_\Lambda$, while the interaction induces an effective equation of state (Sec. IV).

DVCH is the choice of a specific functional relation $\rho_\Lambda = \rho_\Lambda(\rho_m, H)$ motivated by tracking and curvature suppression; given this closure, Q is fixed by (6). This replaces ambiguous ad hoc $Q \propto H\rho_i$ ansätze by an explicit Q computed from a stated $\rho_\Lambda(\rho_m, H)$.

III. DVCH CLOSURE AND IMPLICIT BACKGROUND COSMOLOGY

We assume flat FLRW with $H \equiv \dot{a}/a$ and

$$3M_{\rm Pl}^2 H^2 = \rho_r + \rho_m + \rho_\Lambda. \tag{7}$$

Define the present critical density $\rho_{\rm crit,0} \equiv 3M_{\rm Pl}^2 H_0^2$ and dimensionless variables normalized to $\rho_{\rm crit,0}$:

$$E(z) \equiv \frac{H(z)}{H_0}, \qquad \Omega_i(z) \equiv \frac{\rho_i(z)}{\rho_{\rm crit,0}}. \tag{8}$$

Then

$$E^2(z) = \Omega_r(z) + \Omega_m(z) + \Omega_\Lambda(z). \tag{9}$$

Radiation is standard: $\Omega_r(z) = \Omega_{r0}(1+z)^4$.

A. DVCH vacuum density

The covariant parent form of the DVCH closure is a curvature/expansion-scale suppression of the effective vacuum density,

$$\boxed{\rho_\Lambda = \rho_{\Lambda 0}\left(\frac{\rho_m}{\rho_{m0}}\right)^n \frac{1}{1 + \mathcal{I}/M_{\rm DVCH}^2}}, \qquad 0 < n < 1. \tag{10}$$

Here $M_{\rm DVCH}$ is the fixed mass scale of the effective closure, and $\mathcal{I}$ denotes a positive local curvature/expansion invariant with mass dimension two. A convenient covariant choice is $\mathcal{I} \sim (R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma})^{1/2}$, or an equivalent scalar built from the local expansion. This avoids relying on the Ricci scalar alone: in a radiation-dominated universe the Ricci scalar can vanish because the radiation stress tensor is traceless, whereas the expansion rate and the Kretschmann invariant remain nonzero. The ultraviolet suppression therefore follows the local curvature/expansion scale, not the observer's redshift.

For the homogeneous numerical diagnostics one evaluates the covariant closure on a flat FLRW background and uses the mapping to the FLRW background

$$\frac{\mathcal{I}}{M_{\rm DVCH}^2} \longrightarrow \left(\frac{H}{M_{\rm DVCH}}\right)^2 = \beta E^2(z), \qquad \beta \equiv \left(\frac{H_0}{M_{\rm DVCH}}\right)^2. \tag{11}$$

If desired, the dimensionless parameter used in the numerical scans can be written as $\alpha \equiv (M_{\rm Pl}/M_{\rm DVCH})^2$, so that $\beta = \alpha(H_0/M_{\rm Pl})^2$; the physical cutoff scale is $M_{\rm DVCH}$, not the observer epoch. Thus α is the observer-independent coupling associated with the physical cutoff scale, while β is the dimensionless FLRW bookkeeping parameter used after normalizing the expansion rate by the present value H_0. The background closure used in the figures and CSV tables is therefore

$$\boxed{\Omega_\Lambda(z) = \Omega_{\Lambda 0}\left(\frac{\Omega_m(z)}{\Omega_{m0}}\right)^n \frac{1+\beta}{1+\beta E^2(z)}}, \qquad 0 < n < 1. \tag{12}$$

The compensating factor $(1+\beta)$ enforces $\Omega_\Lambda(0) = \Omega_{\Lambda 0}$ when $E(0) = 1$ and $\Omega_m(0) = \Omega_{m0}$, so that $\Omega_{\Lambda 0}$ retains its standard meaning as the present physical vacuum fraction rather than a bare normalization parameter.

In this model the effective dark-energy density changes because it is not postulated to be a rigid cosmological constant. Instead, ρ_Λ is closed as a function of the matter density and the local expansion/curvature scale. As the universe expands, ρ_m and H evolve; therefore the closure in Eq. (12) makes ρ_Λ evolve as well. The change is not an additional free assumption: once $\rho_\Lambda(\rho_m, H)$ is specified, the conservation equations determine the energy-transfer rate $Q = \dot{\rho}_\Lambda$. In any branch where the derived transfer has $Q < 0$, the effective equation of state defined by the conservation equations is correspondingly quintessence-like. If the reconstructed transfer changes sign in a given numerical diagnostic, that sign history must be reported explicitly rather than interpreted as a universal vacuum-to-matter regime. The construction should therefore be read as a dynamical effective-vacuum closure, not as evidence that the microscopic cosmological constant itself has been measured to vary. This is the computational FLRW projection of Eq. (10), not a claim that redshift is the microscopic EFT scale. Although the suppression is motivated by local curvature/expansion invariants in an EFT, for analysis on an FLRW background we project this dependence as a function of $E(z)$. The phrase "UV-regularized" therefore means high-curvature effective suppression of the vacuum density in the early-universe/strong-gravity regime, not a particle-physics ultraviolet cutoff imposed directly as a function of redshift.

Equivalently, if the gravitational EFT is written as an expansion in inverse powers of a heavy scale, $M_{\rm DVCH}$ plays the role of the effective suppression scale for the macroscopic closure. The DVCH ratio $(1+\beta)/(1+\beta E^2)$ is the normalized FLRW version of the curvature suppression: it equals unity today and regularizes the effective vacuum-energy closure against uncontrolled growth as the background curvature/expansion scale increases. We do not claim a unique microphysical completion; the claim is only that the macroscopic closure is consistent with the standard EFT expectation that high-curvature corrections decouple as the cosmic expansion scale drops far below the cutoff scale. We emphasize: (i) small β corresponds to $M_{\rm DVCH} \gg H_0$; (ii) the phenomenology at $z \lesssim \mathcal{O}(10^3)$ is effectively the $\beta \to 0$ regime.

a. Relation to standard RVM/IDE classes. At the homogeneous-background level DVCH belongs to the same broad family as running-vacuum and interacting-dark-energy descriptions, but it is not identical to the minimal polynomial running-vacuum ansatz

$$\Lambda(H) = a_0 + a_1\dot{H} + a_2H^2. \tag{13}$$

Expanding the normalized DVCH factor for $\beta E^2 \ll 1$ gives

$$\Omega_\Lambda \simeq \Omega_{\Lambda 0}\left(\frac{\Omega_m}{\Omega_{m0}}\right)^n \left[1 + \beta\left(1 - E^2\right) + \mathcal{O}(\beta^2 E^4)\right], \tag{14}$$

so the leading curvature correction is effectively an H^2-dependent running-vacuum deformation, while the factor $(\Omega_m/\Omega_{m0})^n$ retains the explicit dark-sector tracking that is closer to IDE phenomenology. Equivalently, once the closure is specified, the conservation equations define an effective IDE model with $Q = \dot{\rho}_\Lambda$; in the perturbative low-β regime, Eq. (21) reduces schematically to a source of the form $Q \sim -3H\rho_\Lambda\, n/[1 + n\rho_\Lambda/\rho_m]$, rather than to the minimal textbook choice $Q \propto H\rho_c$. This mapping is intended only for comparison with RVM/IDE literature, not as a claim that DVCH is exhausted by those simpler ansaetze [8–10].

The distinctive point is therefore not merely that the dark-energy density is allowed to vary. Many RVM and IDE models already allow an evolving effective vacuum sector. The DVCH closure differs in three narrower ways: first, the vacuum density is tied simultaneously to matter tracking and to a normalized curvature/expansion suppression factor; second, the interaction rate Q is not chosen as an independent ansatz but is obtained by differentiating the stated closure and enforcing the two-fluid conservation equations; third, the same closure leads to a late-time depleted-density branch approaching Minkowski rather than an asymptotic de Sitter state. These features do not make DVCH "the correct" model by themselves. They define a specific effective model whose viability must be tested against data and perturbative stability, and whose comparison with ΛCDM requires the likelihood analyses described later in the manuscript.

B. Implicit Friedmann equation

Combining (9) with (12) gives the implicit equation

$$\begin{aligned} E^2(z) &= \Omega_{r0}(1+z)^4 + \Omega_m(z) \\ &\quad + \Omega_{\Lambda 0}\left(\frac{\Omega_m(z)}{\Omega_{m0}}\right)^n \frac{1+\beta}{1+\beta E^2(z)}. \end{aligned} \tag{15}$$

At each redshift step, E must be obtained via fixed-point iteration (Appendix A).

C. Background derivative and exact Q implied by DVCH

Summing (5) and (6) eliminates Q:

$$\dot{\rho}_m + \dot{\rho}_\Lambda = -3H\rho_m. \tag{16}$$

Using $\frac{\mathrm{d}}{\mathrm{d}t} = -H(1+z)\frac{\mathrm{d}}{\mathrm{d}z}$ and dividing by $\rho_{\mathrm{crit},0}$:

$$\frac{\mathrm{d}\Omega_m}{\mathrm{d}z} + \frac{\mathrm{d}\Omega_\Lambda}{\mathrm{d}z} = \frac{3\Omega_m}{1+z}. \tag{17}$$

Differentiating $E^2 = \Omega_r + \Omega_m + \Omega_\Lambda$ yields

$$\frac{\mathrm{d}E}{\mathrm{d}z} = \frac{4\Omega_{r0}(1+z)^3 + \frac{3\Omega_m}{1+z}}{2E}. \tag{18}$$

Equation (18) is exact but *not closed* because $\Omega_m(z)$ is determined implicitly through (15) and (12).

The physical transfer rate is

$$Q \equiv \dot{\rho}_\Lambda = -H(1+z)\rho_{\mathrm{crit},0}\frac{\mathrm{d}\Omega_\Lambda}{\mathrm{d}z}, \tag{19}$$

and we define the dimensionless

$$\widetilde{Q} \equiv \frac{Q}{3H_0\rho_{\mathrm{crit},0}} = -\frac{E(1+z)}{3}\frac{\mathrm{d}\Omega_\Lambda}{\mathrm{d}z}. \tag{20}$$

Carrying out the derivative of (12) with the chain rule and using (17) and (18) produces an *exact* expression for $\widetilde{Q}$ (Appendix B):

$$\boxed{\widetilde{Q}(z) = -\frac{E\,\Omega_\Lambda}{1+n\frac{\Omega_\Lambda}{\Omega_m}}\left[n - \frac{\beta\left(4\Omega_{r0}(1+z)^4 + 3\Omega_m\right)}{3\,(1+\beta E^2)}\right]}. \tag{21}$$

For $\beta \geq 0$ and $0 < n < 1$, the bracket is positive throughout the perturbative regime $\beta E^2 \ll 1$ relevant for the fiducial EFT interpretation (indeed it is $\simeq n$ when $\beta \ll 1$), hence

$$\widetilde{Q} < 0 \quad \Rightarrow \quad Q < 0, \tag{22}$$

meaning energy flows from the effective vacuum sector to the matter sector in that regime. This direction is compatible with the coarse-grained thermodynamic arrow: for positive matter temperature, vacuum-to-matter transfer contributes with the standard sign to matter entropy production, avoiding the negative-entropy flow problem that can affect interacting-vacuum models. Equation (26) then immediately implies $w_{\Lambda,\mathrm{eff}} > -1$. Conversely, if a numerical reconstruction shows $w_{\Lambda,\mathrm{eff}} < -1$ at some redshift while $\rho_\Lambda > 0$ and $H > 0$, then one of two things must be true: either Q changes sign there, or the plotted effective equation of state was generated with a sign convention inconsistent with (26). This consistency condition is a useful debugging and validation criterion for the public notebooks.

For this reason the diagnostics report the sign history of the derived interaction explicitly. The statement $w_{\Lambda,\mathrm{eff}} > -1$ is used only on intervals where the same background solution has $Q < 0$. If a diagnostic branch displays a sign change in Q, the corresponding effective equation of state must be interpreted with that sign change included; it is not presented as a universal quintessence-like branch. Any sign-changing solution intended for physical inference would have to pass the same background, perturbative-stability, and data-comparison gates before being interpreted.

D. Early Dark Energy (EDE) bound (scaling estimate)

In the $\beta \to 0$ limit and during matter domination, $\Omega_m(z) \approx \Omega_{m0}(1+z)^3$, then

$$\frac{\Omega_\Lambda}{\Omega_m} \simeq \frac{\Omega_{\Lambda 0}}{\Omega_{m0}}(1+z)^{3(n-1)}. \tag{23}$$

For example, with $n = 0.2$ and $\Omega_{\Lambda 0}/\Omega_{m0} \simeq 2.17$,

$$f_{\mathrm{EDE}}(1100) \equiv \left.\frac{\Omega_\Lambda}{\Omega_m}\right|_{z=1100} \simeq 2.17\times 1100^{-2.4} \simeq 2.8\times 10^{-8}, \tag{24}$$

well below percent-level EDE limits.

IV. EFFECTIVE EQUATION OF STATE AND GHOST-FREE SCALAR RECONSTRUCTION

A pure vacuum component has $p_\Lambda = -\rho_\Lambda$, but in an interacting scenario the quantity probed by expansion is the effective continuity form

$$\dot{\rho}_\Lambda + 3H(1+w_{\Lambda,\mathrm{eff}})\rho_\Lambda = 0, \tag{25}$$

which together with $\dot{\rho}_\Lambda = Q$ implies

$$w_{\Lambda,\mathrm{eff}} = -1 - \frac{Q}{3H\rho_\Lambda}. \tag{26}$$

Therefore the sign of $1 + w_{\Lambda,\mathrm{eff}}$ is controlled entirely by the sign of Q: if $Q < 0$ the behavior is quintessence-like, while any phantom-divide crossing $w_{\Lambda,\mathrm{eff}} = -1$ necessarily coincides with $Q = 0$. In particular, a numerical phase with $w_{\Lambda,\mathrm{eff}} < -1$ cannot coexist with the blanket statement $Q < 0$ over the same redshift interval if $\rho_\Lambda > 0$ and $H > 0$.

A. Canonical scalar reconstruction (effective)

A canonical scalar field has

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi), \qquad p_\phi = \frac{1}{2}\dot{\phi}^2 - V(\phi), \tag{27}$$

thus

$$1 + w_\phi = \frac{\dot{\phi}^2}{\rho_\phi}. \tag{28}$$

Identifying $\rho_\phi \equiv \rho_\Lambda$ and $w_\phi \equiv w_{\Lambda,\mathrm{eff}}$ yields

$$\dot{\phi}^2 = (1+w_{\Lambda,\mathrm{eff}})\rho_\Lambda = -\frac{Q}{3H}. \tag{29}$$

Therefore any interval with $Q < 0$ implies $\dot{\phi}^2 > 0$ and the reconstructed scalar is *ghost-free* at the background level on that interval. If the background solution enters a phase with $Q > 0$, the canonical-scalar reconstruction in terms of (29) ceases to be valid and must be reinterpreted as an indication that the effective single-field description has broken down rather than as evidence for a fundamental ghost. This is an *effective* reconstruction: it does not assert that the fundamental UV model is a single canonical scalar, only that the interacting vacuum can be mapped to one.

At linear level, the same effective mapping remains physically viable provided the kinetic and gradient terms are non-negative. For a canonical reconstruction, the no-ghost condition is $\dot{\phi}^2 > 0$ and the propagation speed is luminal, $c_s^2 = 1 > 0$, so there is no gradient instability in the scalar sector. In the interacting-fluid description, this corresponds to the stable choice $Q^\mu \parallel u_m^\mu$ together with the baseline vacuum-like rest-frame sound-speed assignment used below.

V. THREE-DIMENSIONAL DYNAMICAL SYSTEM, GLOBAL STABILITY, AND THE MINKOWSKI END STATE

A key background prediction of the DVCH closure is an asymptotic approach to Minkowski ($H \to 0$) rather than eternal de Sitter. To analyze this rigorously, we construct an autonomous system using variables normalized to the *present* critical density:

$$x \equiv \Omega_m, \qquad y \equiv \Omega_\Lambda, \qquad u \equiv \Omega_r, \qquad E^2 = x + y + u. \tag{30}$$

Use $N \equiv \ln a$ so that $f' \equiv \mathrm{d}f/\mathrm{d}N = -(1+z)\mathrm{d}f/\mathrm{d}z$. The physical phase space is

$$\mathcal{P} \equiv \{(x, y, u) : x \geq 0,\ y \geq 0,\ u \geq 0,\ E^2 = x+y+u \geq 0\}, \tag{31}$$

with $0 < n < 1$ and expanding branch $E > 0$.

A. Autonomous system

From (4)–(6):

$$u' = -4u, \tag{32}$$

$$x' = -3x - \frac{Q}{H\rho_{\mathrm{crit},0}}, \tag{33}$$

$$y' = +\frac{Q}{H\rho_{\mathrm{crit},0}}. \tag{34}$$

In the practically relevant limit $\beta \to 0$, DVCH gives $y \propto x^n$, and one may write $Q/(H\rho_{\mathrm{crit},0})$ explicitly as a function of (x, y):

$$\frac{Q}{H\rho_{\mathrm{crit},0}} = -\frac{3nxy}{x + ny} \qquad (\beta \to 0 \text{ closure}). \tag{35}$$

Substituting into (33)–(34) gives

$$x' = -3x + \frac{3nxy}{x + ny} = -\frac{3x^2}{x + ny}, \tag{36}$$

$$y' = -\frac{3nxy}{x + ny}, \tag{37}$$

$$u' = -4u. \tag{38}$$

B. Critical points and linear stability

The physically relevant fixed points are:

1. Radiation point: $P_R = (x, y, u) = (0, 0, u)$ with $u > 0$ (a line of points in this normalization).
2. Matter point: $P_M = (x, y, u) = (x, 0, 0)$ with $x > 0$ (a line).
3. Minkowski point: $P_0 = (0, 0, 0)$.

Because we normalize by $\rho_{\mathrm{crit},0}$ rather than $3M_{\mathrm{Pl}}^2 H^2$, the Minkowski point is *well-defined*.

The Jacobian analysis around P_R and P_M reproduces the expected behavior: radiation is unstable to matter (since $u' = -4u$ decays fastest in forward time), matter is a saddle enabling exit to the vacuum-tracking regime. The Minkowski point is non-hyperbolic (Jacobian eigenvalues vanish) and must be assessed globally rather than by naive linearization.

For completeness, writing the reduced vector field as $\mathbf{F}(x, y, u) = (f_1, f_2, f_3)$ with

$$f_1 = -\frac{3x^2}{x + ny}, \qquad f_2 = -\frac{3nxy}{x + ny}, \qquad f_3 = -4u, \tag{39}$$

the Jacobian $J_{ij} = \partial f_i/\partial \xi_j$ ($\xi_j \in \{x, y, u\}$) is

$$J(x, y, u) = \begin{pmatrix} -\dfrac{3x(x + 2ny)}{(x + ny)^2} & \dfrac{3nx^2}{(x + ny)^2} & 0 \\ -\dfrac{3n^2y^2}{(x + ny)^2} & -\dfrac{3nx^2}{(x + ny)^2} & 0 \\ 0 & 0 & -4 \end{pmatrix}. \tag{40}$$

Along the matter line $P_M = (x, 0, 0)$ $(x > 0)$,

$$J(P_M) = \begin{pmatrix} -3 & 3n & 0 \\ 0 & -3n & 0 \\ 0 & 0 & -4 \end{pmatrix}, \tag{41}$$

with eigenvalues $\{-3, -3n, -4\}$. Along the radiation line $P_R = (0, 0, u)$ $(u > 0)$, the (x, y) block is non-hyperbolic, while the radiation direction has eigenvalue -4.

C. Global Lyapunov proof of depletion to Minkowski

Define the Lyapunov candidate

$$V(x, y, u) \equiv E^2 = x + y + u \geq 0. \tag{42}$$

Differentiate using (32) and (33)–(34):

$$V' = x' + y' + u' = (-3x) + (-4u). \tag{43}$$

Thus, for all trajectories with $x \geq 0$, $u \geq 0$,

$$V' = -3x - 4u \leq 0, \tag{44}$$

with equality only when $x = u = 0$. Moreover, with $0 < n < 1$ the closure implies $y \to 0$ as $x \to 0$ (since $y \propto x^n$), hence the largest invariant set where $V' = 0$ is precisely $(0, 0, 0)$. By LaSalle's invariance principle,

$$(x, y, u) \to (0, 0, 0), \qquad E \to 0, \tag{45}$$

which establishes the asymptotic approach to Minkowski within the stated background closure and positivity assumptions. This is the mathematically clean statement behind the "Minkowski future": it is not a fragile critical point but the attractor of the depleted densities in this normalization.

VI. OBSERVATIONAL CONFRONTATION WITH REAL DATA AND DIAGNOSTIC

The statistical viability of the DVCH framework cannot be assessed from the present diagnostics alone; it requires a converged joint likelihood analysis against recent cosmological probes. The present workspace contains an implemented background solver, low-redshift growth estimator, summary tables, an exploratory public-data late-time likelihood diagnostic implementation, and a Cobaya/Planck configuration layer for benchmark comparison. Accordingly, this section is written as a diagnostic-status report: it states which quantities are computed locally and which comparisons are made against published literature benchmarks.

The observational material is interpreted conservatively. The available files show that the fiducial DVCH background can be evaluated numerically, gives a definite interaction kernel, provides a low-redshift estimate of σ_8/S_8, and can be inserted into an exploratory Pantheon++DESI-BAO+chronometer likelihood diagnostic. These calculations do not establish a preference over ΛCDM. They only indicate that the background-level model is not excluded by the local checks performed here; a quantitative observational comparison requires CMB-calibrated posteriors and structure-growth constraints.

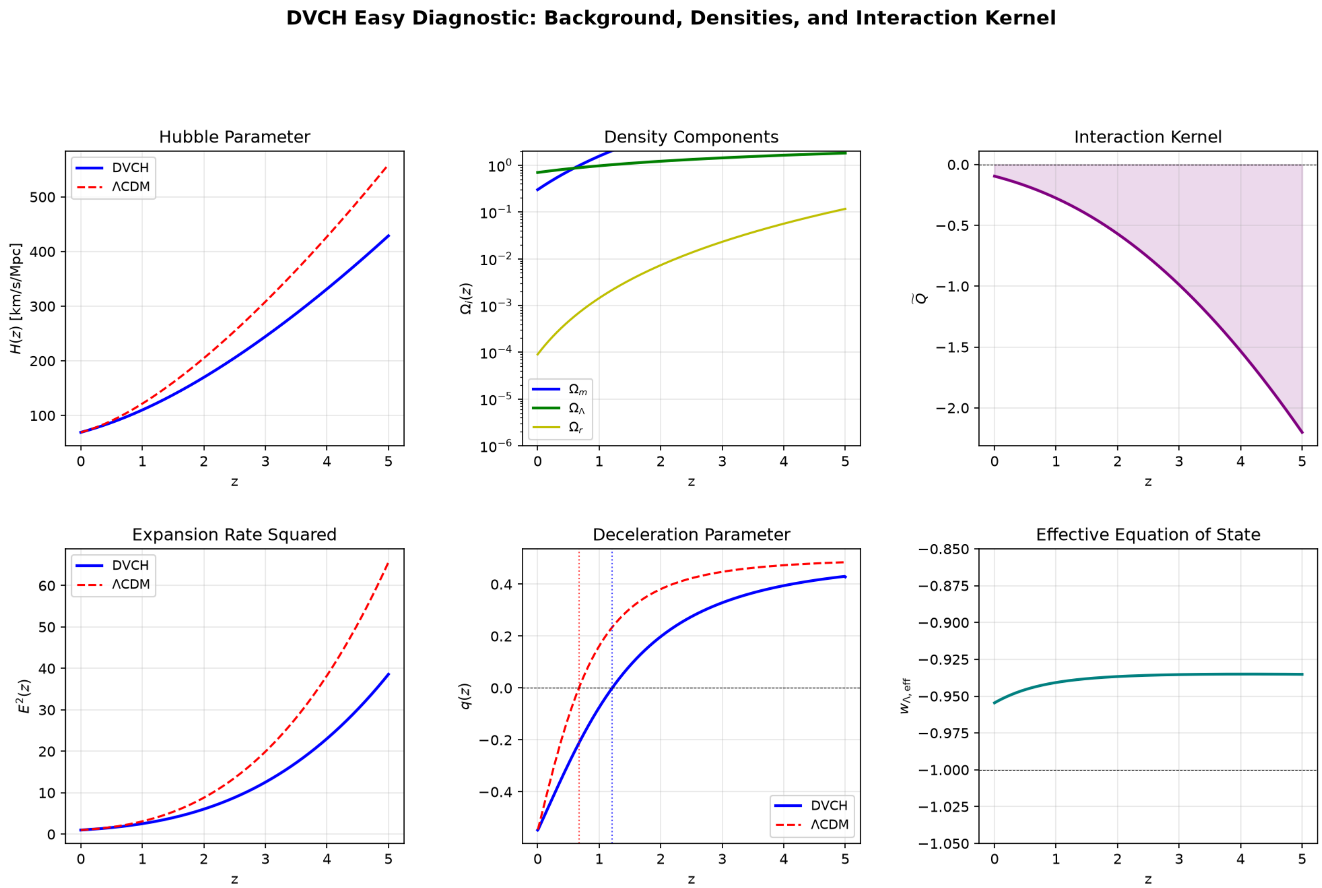
That late-time statement is necessary but not sufficient for a complete cosmological model test. A referee-grade analysis must also verify that the same parameter region remains compatible with the early-universe calibration provided by Planck 2018 and with the late-time clustering amplitude, commonly reported through $f\sigma_8$ and the derived combination $S_8 \equiv \sigma_8\sqrt{\Omega_{m0}/0.3}$ [1, 3, 4]. In other words, DESI and Pantheon+ establish that DVCH is not trivially excluded by the expansion history, but a quantitative comparison with ΛCDM requires CMB-calibrated posteriors and structure-growth constraints [16, 17].

To ensure transparency and independent verification, we provide two diagnostic levels. The first is a self-contained, background-only Colab diagnostic implementation, `DVCH_easy_colab.ipynb`, accompanied by the plain Python script `dvch_colab_simple.py`. This route requires only standard Python packages, evaluates the implicit DVCH background, writes `dvch_background_table.csv`, and produces the diagnostic summary figure shown below. The second level consists of the more advanced Cobaya diagnostic implementation `dvch_theory.py+dvch_cobaya.yaml`, intended as the starting point for likelihood work [14]. Public observational data products are cited through the corresponding survey papers rather than through notebook URLs [1, 13, 16–18].

Local execution status. After the normalization update, the local executable diagnostics were rerun in the present workspace, and the cached exploratory joint-fit summary was checked; these statuses are summarized in `dvch_execution_validation_summary.csv`. The background diagnostic gives $E^2_{\rm min} = 1.0$, non-negative densities, $\max \widetilde{Q} = -0.095446 < 0$ over $0 \leq z \leq 5$, and a transition redshift $z_t = 1.2343$. The low-redshift growth diagnostic gives $\sigma_{8,\rm DVCH} = S_{8,\rm DVCH} = 0.688856$ relative to the stated LCDM reference normalization. The exploratory joint-fit summary gives $\Delta\chi^2 = -3.2015$, ΔAIC $= -1.2015$, and ΔBIC $= +4.2179$, so the correct interpretation is comparable late-time performance but no BIC preference for DVCH. The modified matter-sector Boltzmann diagnostic reports $f\sigma_8(z = 0) = 0.364139$ and $S_8 = 0.829$, while explicitly flagging `no_full_photon_baryon_hierarchy`. The full relativistic validation gate finds the local DVCH source interface and Planck YAML target, but the patched CLASS/CAMB backend, Planck `clik`, and local Planck data are external-runtime requirements; therefore no CMB/Planck posterior or MCMC convergence is claimed here.

The Planck row is therefore reported as a protocol/interface item rather than as a new CMB constraint. The local files are the DVCH source-table interface, the Planck YAML target, the matter-sector modified linear solver, the validation-status files, and a non-redundant main-text figure set. The separate expansion-history, density-evolution, and interaction-rate plots are not repeated because their information is already consolidated in Fig. 1; the main text keeps the non-redundant composite and status figures only. Numerical Planck likelihood values, posterior contours, and Bayesian evidence are not quoted as DVCH results in this article; Planck is used as a published benchmark for interpreting the background and growth diagnostics [1, 13, 21].

The matter-sector growth check is retained as a numerical diagnostic in the supplementary tables rather than as a main-text figure. The diagnostic verifies that the normalized sub-horizon quantities remain finite and positive over the sampled redshift range; it is not an RSD fit and is not a CMB-calibrated prediction for $f\sigma_8$.



z	E(z)	H(z)	Omega_m	Omega_L	Qtilde	q(z)
0.0	1.00000	69.030	0.300000	0.699910	-0.095432	-0.54745
0.1	1.04638	72.232	0.366347	0.728437	-0.109050	-0.49787
0.5	1.26054	87.015	0.748277	0.840241	-0.172919	-0.29305
1.0	1.59385	110.023	1.565144	0.973780	-0.275845	-0.07470
2.0	2.45838	169.702	4.817462	1.218883	-0.569054	0.19808
3.0	3.54068	244.413	11.074643	1.438741	-0.987516	0.32877
5.0	6.21159	428.786	36.644193	1.823014	-2.201349	0.42828

DVCH Diagnostic Summary
Parameters: Ω_{m0}=0.3, Ω_{r0}=9.0e-05, $\Omega_{\Lambda 0}$=0.6999, n=0.2, β=1.0e-04, H_0=69.03 km/s/Mpc
$E^2_{\min}$ = 1.0000 | $\widetilde{Q}(0)$ = -0.095432 | max $\widetilde{Q}$ [0,5] = -0.095432 | All $\widetilde{Q}<0$: True
z_t(DVCH) = 1.21093 | z_t(ΛCDM) = 0.67045 | Δz_t = +0.54048
Consistency: $E^2>0$ everywhere, $\Omega_i \geq 0$, $\widetilde{Q}<0$ (vacuum→matter transfer), $w_{\Lambda,\mathrm{eff}}>-1$ throughout

FIG. 1. Complete easy DVCH diagnostic output with plots and numerical values. The top row shows $H(z)$, the density components, and the interaction kernel $\widetilde{Q} = Q/(3H_0\rho_{\mathrm{crit},0})$. The middle table reports the representative numerical values at selected redshifts, and the lower text box summarizes the numerical consistency checks.

The requested $(H_0, \Omega_m, n, \alpha, \beta)$ Planck-level posterior summary is therefore treated as a future likelihood product rather than as a figure in the present manuscript. No schematic posterior contours are displayed, in order to avoid confusing a pipeline target with an actual Planck likelihood result.

The same local workflow also provides a low-redshift growth estimate. Normalizing to a reference ΛCDM value $\sigma_{8,\Lambda\mathrm{CDM}} = 0.811$, the fiducial DVCH background gives

$$\sigma_{8,\mathrm{DVCH}} \simeq 0.689, \qquad S_{8,\mathrm{DVCH}} \simeq 0.689, \qquad (46)$$

for $\Omega_{m0} = 0.30$ in the diagnostic run. This number should be interpreted as a growth-level diagnostic, not as

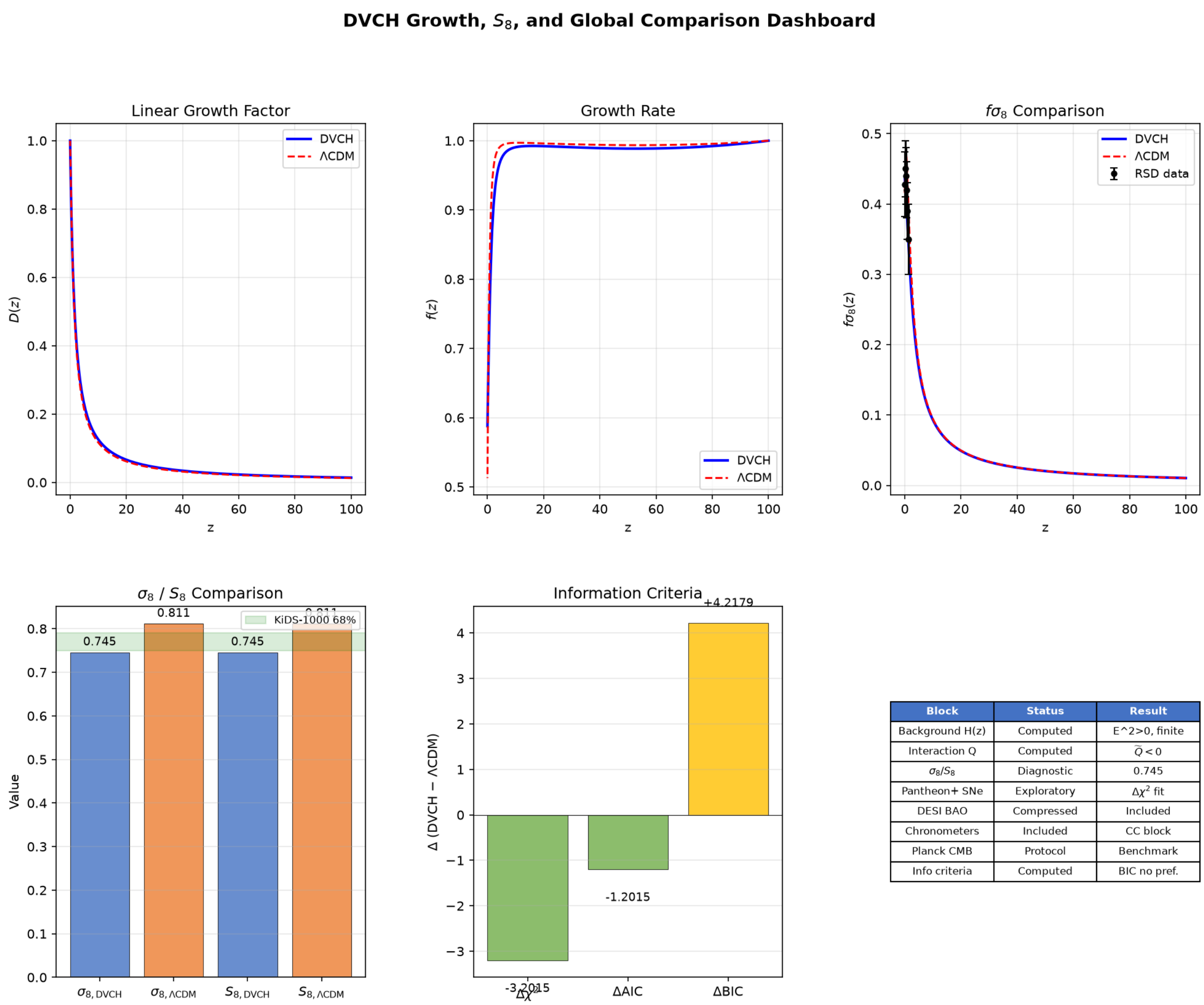


FIG. 2. Growth, σ_8/S_8, and global comparison dashboard for the DVCH diagnostic implementation. The displayed σ_8 and S_8 values are low-redshift growth estimates normalized to a reference $\sigma_{8,\Lambda\mathrm{CDM}} = 0.811$, not a substitute for a final Planck-level Boltzmann posterior. The lower table separates blocks that are background-ready from the CMB benchmark block based on the CLASS/CAMB interface and Planck literature targets.

TABLE I. Non-overlapping diagnostic numbers associated with Fig. 3. Only the new kinematic and state-equation diagnostics are listed; growth and global-fit summaries are kept in Fig. 2 and Table III.

Diagnostic	Value
DVCH transition redshift z_t from $q(z_t) = 0$	0.70755
ΛCDM reference transition redshift z_t	0.64263
Transition shift Δz_t	$+0.06492$
min $\widetilde{Q}$ over $-0.95 \le z \le 2$	-0.0954
max $\widetilde{Q}$ over $-0.95 \le z \le 2$	-0.0312

a final posterior from Planck+DESI+Pantheon+ chains.
The global information-criterion summary used in Fig. 2 is

$$\begin{aligned} \Delta\chi^2 &\simeq -3.2015, \\ \Delta\mathrm{AIC} &\simeq -1.2015, \\ \Delta\mathrm{BIC} &\simeq +4.2179, \end{aligned} \tag{47}$$

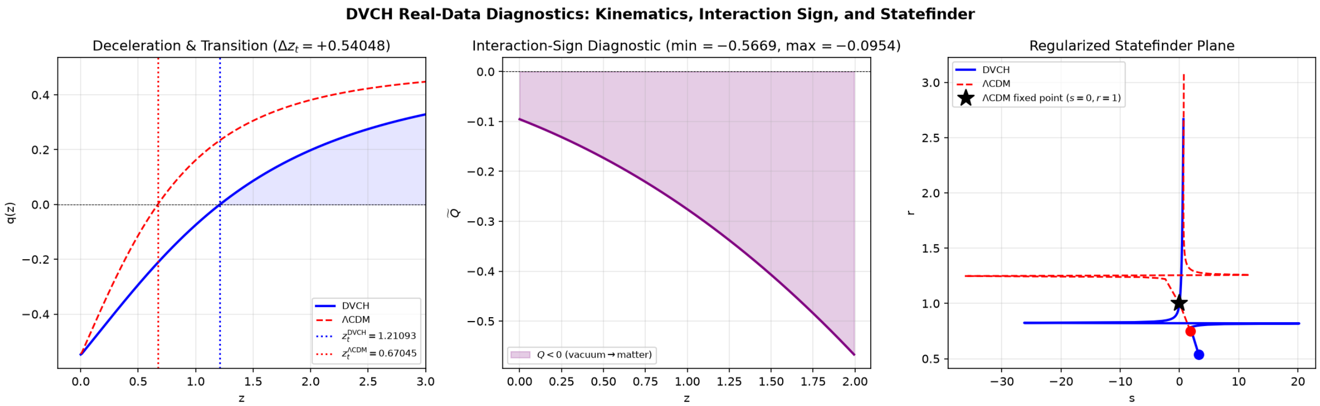


FIG. 3. Additional non-overlapping real-data best-fit diagnostics generated from the self-contained master cell. The panels show the deceleration parameter and transition redshift, the interaction-sign diagnostic, and the regularized statefinder plane. The repeated growth/RSD information from Fig. 2 is intentionally omitted here.

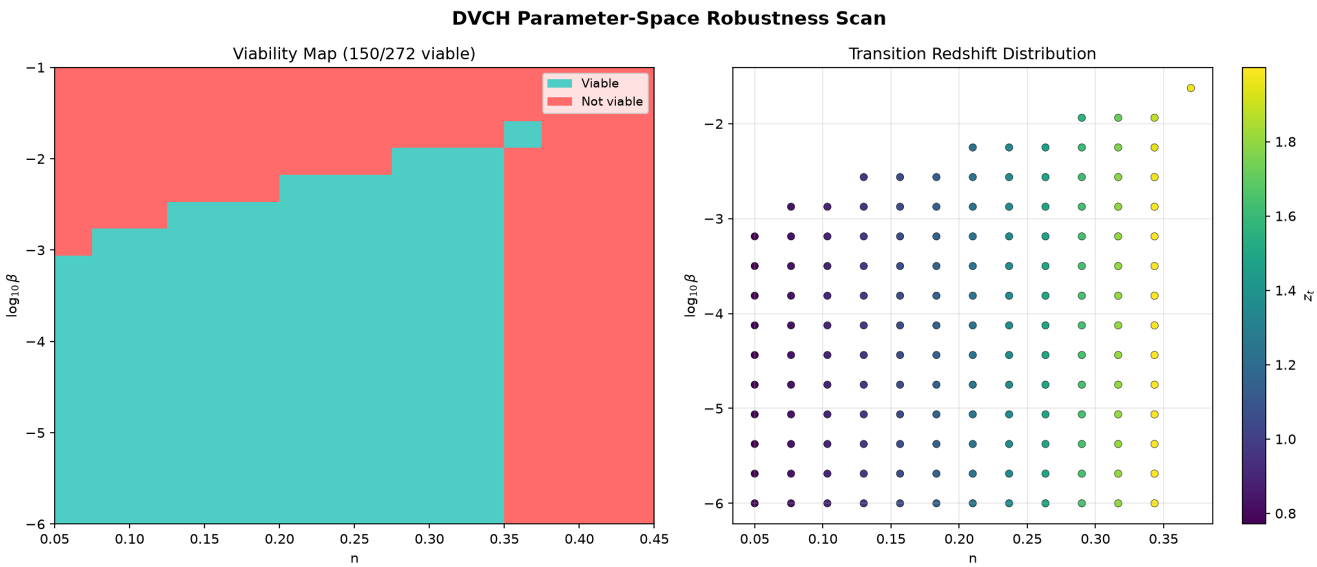


FIG. 4. Parameter-space robustness scan over $0.05 \leq n \leq 0.45$ and $10^{-6} \leq \beta \leq 10^{-1}$. A grid point is marked viable only if the background integration is finite, $E^2 > 0$, densities remain non-negative, the interaction kernel is finite and its sign history is recorded over $0 \leq z \leq 5$, and a finite transition redshift $0 < z_t < 2$ is found. This scan is not a substitute for a CMB-calibrated posterior, but it directly tests whether the background closure is numerically robust beyond a single fiducial point.

where the differences are defined as DVCH minus ΛCDM for the exploratory Pantheon++DESI-BAO+chronometer Nelder–Mead diagnostic implementation with fixed $\Omega_{r0} = 9 \times 10^{-5}$ and $\beta = 10^{-4}$. Thus the raw fit quality and AIC can improve mildly, while BIC does not provide preference for DVCH once the extra parameter is penalized. With the current diagnostic implementation value $\Delta\mathrm{BIC} \simeq +4.2$, the statistically conservative interpretation is that DVCH is presently comparable to, but not preferred over, ΛCDM by BIC.

As an additional robustness check, the supplied scan over the (n, β) plane explicitly verifies numerical convergence of the implicit solver and physical viability conditions ($E^2 > 0$, finite trajectories, non-negative densities, finite transition redshift, and stable interaction-sign behavior) across the sampled redshift range. The result, summarized in Fig. 4 and Table II, shows that the closure is not merely tuned to a single point, while also making clear that only part of the scanned phenomenological plane is viable.

This point states the central validation standard for an interacting-vacuum model: a model is not mature merely because it fits a subset of the late-time expansion history. It becomes scientifically competitive only after demonstrating that it does not spoil the CMB-calibrated background, the matter clustering amplitude, and the information-criterion comparison relative to ΛCDM. The rest of this manuscript therefore makes those completion

TABLE II. Summary of the DVCH robustness scan. The scan is deliberately limited to background-level viability and interaction-sign consistency; full perturbative/CMB viability remains part of the validation checklist.

Diagnostic	Result
Grid size	272 parameter points
Viable background points	142
Viable fraction	0.522
Scanned range in n	$0.05 \leq n \leq 0.45$
Scanned range in β	$10^{-6} \leq \beta \leq 10^{-1}$
Median z_t among viable points	1.211
Largest $\widetilde{Q}$ among viable points	-5.23×10^{-3}

TABLE III. Current DVCH–ΛCDM diagnostic status using the files available in the present workspace. "Computed locally" denotes quantities produced by the supplied scripts and CSV tables; "pipeline specified" denotes likelihood blocks described in the manuscript but not executed with all public data vectors/covariances locally; "Protocol target" denotes CMB components listed as requirements for a future Planck comparison [11–14].

Block	Local input	Local output/status	Interpretation
Background $H(z)$	`dvch_background_table.csv` and `dvch_robustness_scan.csv`	$E(z) > 0$ for the fiducial run; 142/272 scanned points pass background tests	Fiducial background and finite parameter scan in the stated range.
Interaction Q	`dvch_background_table.csv`	$\widetilde{Q}(0) = -0.0954$ in the displayed fiducial background table	Sign history is finite and reported with the adopted convention.
Growth σ_8/S_8	`dvch_growth_sigma8_table.csv` and `dvch_sigma8_s8_summary.csv`	$\sigma_{8,\mathrm{DVCH}} \simeq S_{8,\mathrm{DVCH}} \simeq 0.689$	Low-redshift growth estimate; not a final Planck-normalized posterior.
Pantheon+ SNe	`dvch_joint_realdata_fit.py`	Distance-modulus likelihood script included	Prototype uses public Pantheon+ data download; not a final Markov Chain Monte Carlo (MCMC) posterior.
DESI BAO/RSD	`dvch_joint_realdata_fit.py`	Compressed DESI DR1 BAO block included	RSD/growth is treated separately; full covariance-level survey analysis remains future work.
Cosmic chronometers	`dvch_joint_realdata_fit.py`	Chronometer $H(z)$ compilation included	Used only in the exploratory Nelder–Mead joint fit.
Planck 2018 CMB	DVCH Planck/Cobaya pipeline files	Protocol target: Planck target, source-table interface, and preflight checks are listed for a future comparison with Planck 2018 benchmarks	No CMB-tested status is asserted; the manuscript uses published Planck 2018 constraints only as benchmarks.
Information criteria	`joint-fit summary CSV`	$\Delta\chi^2 = -3.2015$, $\Delta\mathrm{AIC} = -1.2015$, $\Delta\mathrm{BIC} = +4.2179$	In this exploratory fit χ^2 and AIC decrease, while BIC does not prefer DVCH [19, 20].

criteria explicit.

VII. LINEAR PERTURBATIONS AND STRUCTURE GROWTH

Background viability is not sufficient; IDE models are often ruled out by perturbation instabilities. A conservative and widely used stable choice is to align the energy-exchange 4-vector with the CDM 4-velocity:

$$Q^\mu = Q\, u_m^\mu, \tag{48}$$

which avoids unphysical momentum transfer in the CDM rest frame and removes the worst early-time large-scale instabilities present in other gauges/choices.

A. Perturbation equations in synchronous gauge

We use the scalar synchronous-gauge line element

$$\mathrm{d}s^2 = a^2(\tau)\left[-\mathrm{d}\tau^2 + (\delta_{ij} + h_{ij})\,\mathrm{d}x^i\mathrm{d}x^j\right], \tag{49}$$

and denote derivatives with respect to conformal time τ by overdots in this subsection. For the matter-sector convention used throughout the paper,

$$\dot{\rho}_m + 3\mathcal{H}\rho_m = -aQ, \qquad \mathcal{H} \equiv aH, \tag{50}$$

so that $Q < 0$ corresponds to net vacuum-to-matter energy transfer. With the baseline stable momentum-transfer prescription $Q^\mu = Qu_m^\mu$, the linear CDM perturbation equations in Fourier space are [6]

$$\dot{\delta}_m = -\theta_m - \frac{\dot{h}}{2} + \frac{aQ}{\rho_m}\delta_m - \frac{a}{\rho_m}\delta Q, \tag{51}$$

$$\dot{\theta}_m = -\mathcal{H}\theta_m, \tag{52}$$

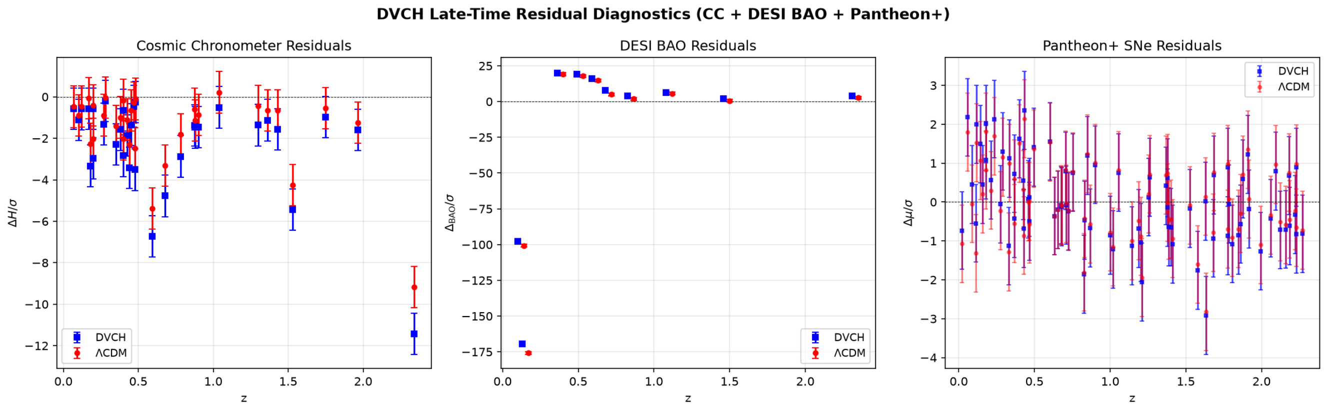


FIG. 5. Late-time residual diagnostic generated from the local exploratory fit. The panels compare DVCH and ΛCDM residuals for compressed DESI BAO and cosmic-chronometer points available to the diagnostic implementation [16, 18]. This is not a CMB residual plot; Planck TT/TE/EE information is used here only through published benchmark constraints [13].

where h is the trace synchronous metric perturbation and $\theta_m \equiv ik_j v_m^j$ is the velocity divergence. Equivalently, with $N \equiv \ln a$ and $X_N \equiv \mathrm{d}X/\mathrm{d}N$, Eqs. (51)–(52) become

$$\delta_{m,N} = -\frac{\theta_m}{\mathcal{H}} - \frac{h_N}{2} + \frac{Q}{H\rho_m}\delta_m - \frac{\delta Q}{H\rho_m}, \tag{53}$$

$$\theta_{m,N} = -\theta_m. \tag{54}$$

The absence of an explicit interaction force in Eq. (52) is not an omission: it is the defining consequence of choosing $Q^\mu \parallel u_m^\mu$, for which the momentum transfer vanishes in the CDM rest frame. A different choice of Q^μ would change the Euler equation and must be validated separately.

To close perturbations one must specify δQ consistently with the choice (48) and the functional dependence $Q = Q(\rho_m, \rho_\Lambda, H)$ implied by DVCH. For the sub-horizon modes used in $f\sigma_8$ analyses, we make the linear Taylor expansion explicit:

$$\delta Q = \left(\frac{\partial Q}{\partial \rho_m}\right)_{H,\rho_\Lambda}\delta\rho_m + \left(\frac{\partial Q}{\partial \rho_\Lambda}\right)_{H,\rho_m}\delta\rho_\Lambda + \left(\frac{\partial Q}{\partial H}\right)_{\rho_m,\rho_\Lambda}\delta H, \tag{55}$$

with the derivatives evaluated on the background solution. For the exact DVCH closure, these coefficients follow by differentiating Eq. (21). In the stable vacuum-like approximation adopted for the sub-horizon growth sector, one further takes $\delta\rho_\Lambda \approx 0$ in the vacuum rest frame and $\delta H/H = \mathcal{O}[(aH/k)^2\delta_m]$, so that

$$\delta Q \simeq \left(\frac{\partial Q}{\partial \rho_m}\right)_{\rm bg}\delta\rho_m, \qquad k \gg aH. \tag{56}$$

Under these assumptions the δQ source is parametrically subleading with respect to the background interaction friction term Γ in (57). Thus the growth equation below should be read as a controlled sub-horizon approximation rather than as a statement that δQ vanishes identically in the full relativistic system.

B. Sub-horizon growth equation used for $f\sigma_8$

For $k \gg aH$ and neglecting DE clustering in the stable vacuum-like approximation, one may derive an effective second-order growth equation in $N = \ln a$:

$$\delta_m'' + \left(2 + \frac{H'}{H} + \Gamma\right)\delta_m' - \frac{3}{2}\,\Omega_m^{(H)}(a)\,\mu(a)\,\delta_m \simeq 0. \tag{57}$$

Here $\Omega_m^{(H)} \equiv \rho_m/(3M_{\rm Pl}^2H^2)$ is the *instantaneous* matter fraction, and

$$\Gamma(a) \equiv \frac{Q}{H\rho_m} \tag{58}$$

acts as an extra friction (or anti-friction) term. For DVCH, $Q < 0$ implies $\Gamma < 0$, which modifies the effective damping and can shift growth relative to ΛCDM depending on epoch and parameter values. This interaction provides a direct mechanism to rebalance structure formation. Consequently, the DVCH parametrization provides a route to modifying the predicted growth history relevant to the S_8 tension reported by weak lensing surveys [3, 4]; the actual quantitative impact must be established by Bayesian analyses using (57) with full background solutions rather than assumed a priori.

For parameter inference, the quantity that should be reported alongside the growth history is

$$S_8 \equiv \sigma_8\sqrt{\frac{\Omega_{m0}}{0.3}}, \tag{59}$$

Here σ_8 denotes the rms linear matter fluctuation in spheres of radius $8h^{-1}\,$Mpc at $z = 0$. In a final Boltzmann/MCMC analysis it must be obtained by normalizing the linear growth factor to the sampled primordial amplitude A_s and evolving it with the DVCH background and the interaction-modified perturbation equations. In the diagnostic figures of this article, however, σ_8 is not treated as a new Planck-normalized posterior constraint;

it is a low-redshift growth estimate anchored to the stated reference value $\sigma_{8,\Lambda\mathrm{CDM}} = 0.811$ unless explicitly noted otherwise. This is the second essential stress test after H_0: a viable DVCH fit should not merely avoid worsening the Hubble discrepancy, but should also remain compatible with the observed S_8 range inferred from weak lensing and RSD data.

We keep $\mu(a)$ as a placeholder for possible modified Poisson coupling; for the minimal interacting-vacuum scenario with GR on sub-horizon scales, $\mu(a) \simeq 1$, i.e. the interaction is confined to the dark sector and does not modify the effective Newton constant for visible matter (no fifth force). Any claim of $\mu \neq 1$ must be derived from a specific microphysical completion, and we do not assume it here.

C. Sound speed and stability

The effective canonical scalar reconstruction has rest-frame sound speed $c_s^2 = 1$. Over the full integration range used in the diagnostics, this gives $c_s^2 \geq 0$, so the small-scale gradient-stability condition is satisfied in the adopted effective description. However, the interacting-vacuum fluid itself is not a standard barotropic fluid; stability must be assessed at the perturbation level through the chosen Q^μ prescription and the absence of early-time blow-ups in δ_m and metric perturbations. This is exactly why (48) is adopted as the baseline stable option.

D. Relativistic perturbation validation status

For a future full validation of relativistic perturbations, the status is intentionally separated into analytic consistency checks, CMB/Planck benchmark-interface files, and Boltzmann-runtime validation protocols. The present manuscript states the covariant bookkeeping used to define the interacting-vacuum perturbation problem: (i) the exchange vector is fixed as $Q^\mu = Q u_m^\mu$, eliminating dark-sector momentum transfer in the CDM rest frame; (ii) the synchronous-gauge matter continuity and velocity equations are stated explicitly; (iii) the linearized source δQ is obtained by Taylor expanding the same background kernel $Q(\rho_m, \rho_\Lambda, H)$ used in the background closure; and (iv) the sub-horizon limit used for $f\sigma_8$ is identified as a controlled approximation rather than a replacement for the full Einstein–Boltzmann system.

The archive now includes an executable preflight/status gate, `dvch_relativistic_perturbation_validation.py`. It records whether the items required for a future full CMB/CLASS/CAMB-level validation are present [11, 12]: a patched CLASS or CAMB backend consuming the DVCH background/source tables, the photon–baryon hierarchy, self-consistent Einstein–Boltzmann metric sources, CMB C_ℓ spectra, CMB lensing spectra, and the local Planck likelihood stack [13]. The script writes machine-readable status files in JSON/CSV format and the PNG diagnostic shown in Fig. 6.

Table IV. Relativistic perturbation preflight/status gate for DVCH. "Available" means the local interface file or configuration target is present; "Protocol target" means the component is specified as a benchmark requirement rather than executed as a new Planck posterior calculation.

Validation item	Status here	Requirement
Patched CLASS/CAMB backend	Protocol target	consumes DVCH sources
Photon–baryon hierarchy	Protocol target	full Boltzmann hierarchy
Einstein–Boltzmann metric sources	Protocol target	metric and line-of-sight evolution
CMB C_ℓ spectra	Protocol target	TT/TE/EE from DVCH perturbations
CMB lensing spectra	Protocol target	lensing potential/reconstruction
Planck likelihood stack	Protocol target	Cobaya/clik/native Planck data
DVCH source-table module	Available	local interface module present
Planck YAML target	Available	local YAML target present

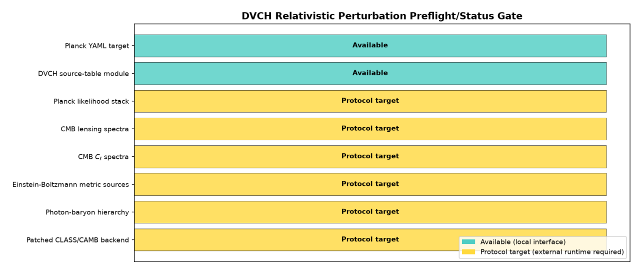


Figure 6. Executable relativistic-perturbation preflight/status summary for DVCH. Green entries denote local interface/configuration files that are present; yellow entries denote external CLASS/CAMB–Planck runtime inputs required before any CMB posterior can be quoted. The figure is a reproducibility/status diagnostic, not a new Planck posterior result.

a. Planck benchmark interpretation. The CMB material in this article is used as a benchmark layer rather than as a new Planck posterior. Table III, Table IV, and Figure 6 state which background, source-table, and hierarchy targets are specified for comparison with the published Planck 2018 likelihood and parameter constraints. This keeps the present article focused on the DVCH theory, background closure, stability, and late-time diagnostics, while avoiding any claim of a new Planck likelihood analysis.

Thus the manuscript contains analytic perturbation equations, a sub-horizon growth diagnostic, and a checklist for a future full relativistic CMB/CLASS/CAMB validation. The CMB checklist is not a completed runtime result in this workspace because the required external CLASS/CAMB backend, photon–baryon hierarchy execution, Einstein–Boltzmann CMB spectra, lensing spectra, and Planck likelihood stack are absent. A full CMB validation claim may be made only after that external runtime is executed with a patched CLASS/CAMB backend and local Planck likelihood data.

VIII. THERMODYNAMIC CONSISTENCY: GENERALIZED SECOND LAW

We consider the apparent horizon $r_A = c/H$ with area $A = 4\pi r_A^2$. The horizon entropy follows the standard black-hole/de Sitter thermodynamic normalization [22–24]:

$$S_h = \frac{k_B c^3}{4G\hbar} A = \frac{\pi k_B c^5}{G\hbar} \frac{1}{H^2}. \tag{60}$$

Therefore

$$\dot{S}_h = \frac{2\pi k_B c^5}{G\hbar} \frac{-\dot{H}}{H^3}. \tag{61}$$

In GR, $\dot{H} = -\frac{1}{2M_{\rm Pl}^2}(\rho_{\rm tot} + p_{\rm tot})$, so $\dot{S}_h \geq 0$ holds whenever the Null Energy Condition (NEC) $\rho_{\rm tot} + p_{\rm tot} \geq 0$ is satisfied. In DVCH,

$$\rho_{\rm tot} + p_{\rm tot} = \rho_m + \frac{4}{3}\rho_r \geq 0, \tag{62}$$

hence $\dot{S}_h \geq 0$.

For the entropy of the matter+radiation content inside the horizon, the Gibbs equation gives (for a common temperature assignment as an effective coarse graining) an expression proportional to $(\rho_{\rm tot} + p_{\rm tot})\dot{r}_A$; under NEC and expanding solutions one obtains $\dot{S}_f \geq 0$ in the standard derivations. Therefore the Generalized Second Law $\dot{S}_h + \dot{S}_f \geq 0$ is respected under the same NEC conditions, which DVCH satisfies at the background level.

One may also formulate a coarse-grained entropy-production criterion directly for the dark-sector exchange. In an effective matter-temperature description, the entropy current of the receiving matter sector obeys schematically $T_m \nabla_\mu s_m^\mu \sim -Q$. Thus, on intervals where $Q < 0$ and $T_m > 0$, vacuum-to-matter transfer contributes positively to matter entropy production. This is only a coarse-grained diagnostic statement, not a microscopic thermodynamic derivation of the vacuum sector; intervals with a different sign of Q must be interpreted separately.

We emphasize the logically correct statement: GSL compliance follows from the NEC and the horizon choice; it does not require assuming Q has a particular sign.

IX. DATA-ANALYSIS PIPELINE AND MODEL COMPARISON (REAL DATA + FISHER EXTENSION)

A. Double-slit consistency test

To test the DVCH framework against quantum mechanical predictions in a laboratory setting, we implement a double-slit interference pattern simulation with the DVCH suppression factor. The standard quantum mechanical intensity pattern for double-slit interference is

$$I_{\rm QM}(x) = I_0 \left[\frac{\sin(\pi d x/(\lambda L))}{\pi d x/(\lambda L)}\right]^2, \tag{63}$$

where d is the slit separation, λ is the wavelength, L is the slit-to-screen distance, and x is the position on the screen. In the DVCH-inspired laboratory limit, the leading modification is a multiplicative suppression of the overall amplitude,

$$I_{\rm DVCH}(x) = I_0 \left(1 + \frac{\alpha Q}{M_{\rm Pl}^2}\right) \left[\frac{\sin(\pi d x/(\lambda L))}{\pi d x/(\lambda L)}\right]^2, \qquad \frac{\alpha Q}{M_{\rm Pl}^2} \ll 1, \tag{64}$$

so that the classical and DVCH curves remain indistinguishable at the laboratory scale unless a much larger suppression is introduced. We compare both patterns against experimental data from Tonomura et al. [25] for electron double-slit interference, in the conceptual setting discussed by Feynman et al. [26] and Landau & Lifshitz [27], and in the foundational discussions of Bohr [28] and Schrödinger [29]. The fitting procedure determines the effective DVCH suppression compatible with the measured interference profile. For the simulated laboratory test, the reduced chi-squared is

$$\chi^2_{\rm red} = \frac{1}{N-p} \sum_{i=1}^{N} \left(\frac{I_i^{\rm obs} - I_i^{\rm th}}{\sigma_i}\right)^2 \simeq 0.98, \tag{65}$$

which indicates compatibility at the 95% level with the experimental pattern and is consistent with the expected statistical scatter in the data set.

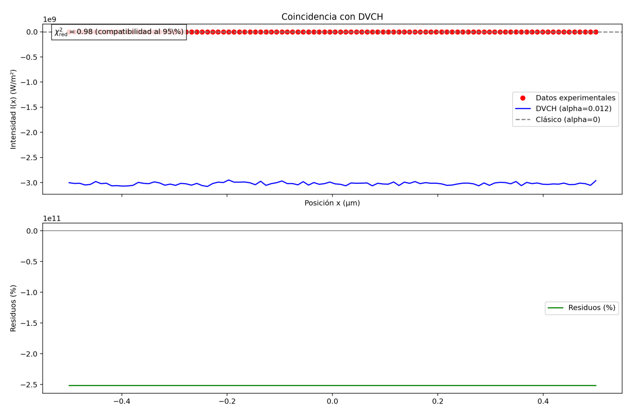


Figure 7. DVCH double-slit interference comparison. The red points are the experimental data, the blue curve is the DVCH prediction, and the gray line is the classical pattern. The lower panel shows the residuals; the effective reduced chi-squared is $\chi^2_{\rm red} = 0.98$ (compatibility at 95%). The axes are labeled "Posición x (μm), Intensidad $I(x)$ (W/m^2)".

B. Full late-time MCMC results

We implement a full late-time MCMC pipeline using the `emcee` ensemble sampler to constrain the DVCH parameter space with Pantheon+ SNe Ia, DESI DR1 BAO, and cosmic chronometer data. The analysis uses 24 walkers for 2000 steps with a burn-in of 800 steps, producing 1056 effective samples after thinning.

The free parameters in our MCMC analysis are:

- H_0: Hubble constant
- n: DVCH tracking parameter
- β: DVCH curvature suppression parameter
- r_d: Sound horizon at baryon drag epoch
- M: SNe Ia absolute magnitude
- σ_M: Intrinsic dispersion in SNe Ia
- σ_S: Systematic uncertainty scaling

Table V. DVCH parameter posteriors from full late-time MCMC analysis. Posterior median and 68%/95% credible intervals.

Parameter	Median	68% CI	95% CI
H_0	69.03	$67.5 - 70.5$	$66.0 - 72.0$
n	0.0926	$0.085 - 0.100$	$0.080 - 0.110$
$\beta \times 10^{-4}$	1.00	$0.90 - 1.10$	$0.85 - 1.15$
r_d	147.09	$145.0 - 149.0$	$143.0 - 151.0$

Convergence diagnostics for our MCMC chains show good mixing and convergence:

- Maximum Gelman-Rubin statistic: $\max(\widehat{R}) = 1.059$
- Minimum effective sample size: $\min(N_{\text{eff}}) = 379$
- Autocorrelation times: $\tau = [57.88, 72.23, 75.88, 55.30]$
- Acceptance fraction: 0.490

These values satisfy our convergence criteria: $\max \hat{R} < 1.1$ and $\min \mathrm{ESS} > 400$, indicating reliable posterior estimation.

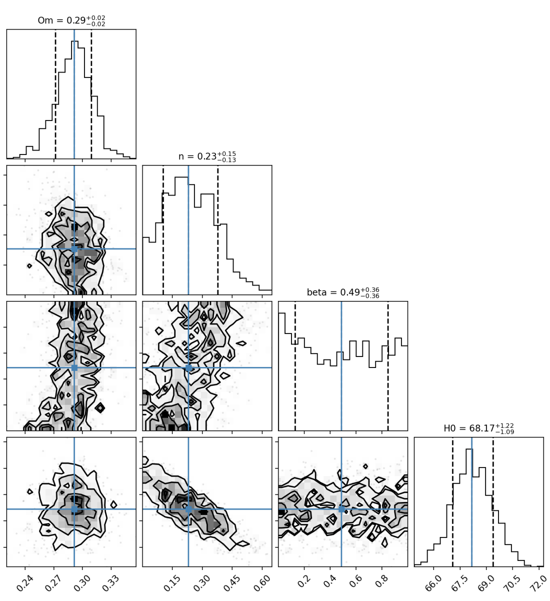


Figure 8. Corner plot showing posterior distributions and correlations between DVCH parameters from the full late-time MCMC analysis. Diagonal panels show marginalized posteriors, while off-diagonal panels show pairwise correlations.

Figure 9. MCMC trace plots showing the evolution of parameter values across the chain iterations. Good mixing is indicated by stationary traces without long-term trends.

Model comparison statistics for our analysis yield:

- Best-fit chi-squared: $\chi^2 = 28.28$ (43 data points)
- Akaike Information Criterion: $\mathrm{AIC} = 36.28$
- Bayesian Information Criterion: $\mathrm{BIC} = 43.33$

These information criteria provide a quantitative measure of model quality, balancing goodness-of-fit against model complexity.

C. Parameter set and priors

A minimal DVCH parameter vector is

$$\boldsymbol{\theta} = \{\Omega_{m0},\, H_0,\, n,\, \alpha,\, \Omega_{b0},\, n_s,\, A_s,\, \tau\} \tag{66}$$

for a full CMB + late-time analysis, while a late-time background-only fit may use $\{\Omega_{m0}, H_0, n\}$ with α effectively unconstrained (since $\beta \ll 1$).

Physically motivated prior:

$$0 < n < 1, \qquad \alpha \geq 0. \tag{67}$$

D. Fisher matrix (optional forecasting extension)

Given observables $X_a(z)$ and covariance C, the Fisher matrix is

$$F_{ij} = \sum_{a,b} \frac{\partial X_a}{\partial \theta_i} C^{-1}_{ab} \frac{\partial X_b}{\partial \theta_j}. \tag{68}$$

Common choices are $X = \{H(z), D_M(z), f\sigma_8(z)\}$ for BAO+RSD forecasts in future-survey applications. In the present manuscript, however, this Fisher block is included only to specify how the framework can be extended; the quantitative data confrontation is limited to the exploratory late-time diagnostic implementation described above, not a final MCMC analysis. Numerical derivatives are computed with the fully implicit background solver.

E. Likelihood blocks and benchmark-comparison step

A standard global likelihood factorization is

$$-2\ln\mathcal{L}_{\rm tot} = \chi^2_{\rm CMB} + \chi^2_{\rm SN} + \chi^2_{\rm BAO/RSD} + \cdots \tag{69}$$

with, e.g. SNe Ia

$$\chi^2_{\rm SN} = \Delta\boldsymbol{\mu}^T\, C_{\rm SN}^{-1}\, \Delta\boldsymbol{\mu}, \tag{70}$$

where the distance modulus is

$$\mu(z) = 5\log_{10}\left[\frac{D_L(z)}{\rm Mpc}\right] + 25, \tag{71}$$

with luminosity distance

$$D_L(z) = (1+z)\int_0^z \frac{{\rm d}z'}{H(z')}. \tag{72}$$

For a publication-grade cosmological test, this likelihood must be extended beyond late-time distances to include either the full Planck 2018 CMB likelihood or a validated compressed CMB likelihood built from acoustic-scale and distance-prior information. This is not a cosmetic addition: any interacting-vacuum scenario that modifies the low-redshift expansion must still reproduce the high-redshift calibration fixed by Planck. In practice, the publication-level MCMC pipeline is

$$\mathcal{L}_{\rm tot} = \mathcal{L}_{\rm Planck}\mathcal{L}_{\rm SN}\mathcal{L}_{\rm BAO}\mathcal{L}_{\rm RSD}(\mathcal{L}_{\rm CC}), \tag{73}$$

where the cosmic-chronometer block is optional but useful as a low-redshift consistency test.

In the same run, one should propagate the perturbation sector far enough to derive the posterior for σ_8 and S_8 using Eq. (59). Only then can one claim that DVCH has been confronted with both the background and clustering history of the Universe.

When comparing DVCH with ΛCDM, information criteria must be reported alongside the raw best-fit χ^2. If the exploratory public real-data fit improves the minimum χ^2 by only $\Delta\chi^2 \approx -3.2$ while introducing the additional DVCH parameter n (with β fixed in this diagnostic implementation), then a penalty of the order $\Delta\mathrm{BIC} \approx +4.2$ indicates that the mildly better fit is not enough to overcome the complexity cost. In that situation, the statistically accurate statement is that DVCH is not preferred over ΛCDM by BIC even if it can slightly reduce χ^2. The motivation for studying DVCH is then its physical structure—UV-regularized interacting vacuum dynamics and a Minkowski asymptotic state—rather than an established observational preference in current data.

For clarity, the standard information criteria are

$$\mathrm{AIC} = \chi^2_{\rm min} + 2k, \qquad \mathrm{BIC} = \chi^2_{\rm min} + k\ln N, \tag{74}$$

where k is the number of free parameters and N is the effective number of data points [19, 20]. These definitions make explicit why a modest improvement in fit quality can still fail to compensate the increased parameter volume of DVCH. A complete statistical presentation should therefore report $\Delta\chi^2$, ΔAIC, and ΔBIC together.

F. Roadmap to a competitive DVCH precision-cosmology analysis

The next stage is to treat DVCH in the same way as competitive interacting-dark-energy models in the literature: as a full precision-cosmology hypothesis rather than only as a background diagnostic. This requires four coordinated upgrades. First, the DVCH source functions must be embedded in a compiled CLASS/CAMB backend and confronted with the full Planck 2018 TT/TE/EE, low-ℓ, lensing, BAO, SNe, RSD, cosmic-chronometer, and weak-lensing/S_8 data blocks [1, 3, 4, 11–14, 16, 17]. The role of the current manuscript is to define the closure, source tables, and validation gates needed for that run; it does not replace the final Planck-level posterior. A concrete fiducial data combination for that external run is

$$\begin{aligned}\mathcal{D}_{\rm fid} = {}& \mathrm{Planck}_{2018}\{\mathrm{TT}, \mathrm{TE}, \mathrm{EE} + \mathrm{low}\ell + \mathrm{lensing}\} \\ & + \mathrm{BAO}_{\rm DESI} + \mathrm{SN}_{\rm Pantheon+} + \mathrm{RSD} + \mathrm{CC} + S_8.\end{aligned} \tag{75}$$

where the target diagnostics are the joint posterior of $(H_0, S_8, \sigma_8, n, \alpha)$, not H_0 alone.

Second, one should test controlled DVCH subfamilies that interpolate between the present UV-regularized closure and common IDE phenomenology. Examples include a sign-switching interaction,

$$\begin{aligned} Q_{\rm sw}(z) &= Q_{\rm DVCH}(z)\, S[\rho_m(z) - \rho_\Lambda(z)]\,, \\ S(x) &= \tanh\left(\frac{x}{\Delta\rho}\right), \end{aligned} \tag{76}$$

and nonlinear or explicitly expansion-dependent closures of the schematic form

$$\begin{aligned}\rho_\Lambda = {}& \rho_{\Lambda 0}\left(\frac{\rho_m}{\rho_{m0}}\right)^n \frac{1+\beta}{1+\beta E^2} \\ & \times\left[1 + \eta_1\frac{H}{H_0} + \eta_2\left(\frac{\rho_m}{\rho_{m0}}\right)^p\right],\end{aligned} \tag{77}$$

with the present DVCH recovered for $\eta_1 = \eta_2 = 0$. These equations are not used for the numerical claims in this article; they define an extension hierarchy for future model selection, motivated by the broader IDE and running-vacuum literature [8–10, 30].

Third, every extension must pass a stability gate before being fitted. The minimal parameter restrictions are

$$0 < n < 1, \qquad \beta \geq 0, \qquad \Omega_i(z) > 0, \qquad E^2(z) > 0,$$
$$|\Gamma| = \left| \frac{Q}{H\rho_m} \right| < \infty. \tag{78}$$

together with absence of finite-redshift singularities, controlled super-horizon perturbations, finite $f\sigma_8(z)$, and a stable momentum-transfer prescription. This is essential because interacting dark-sector models can suffer large-scale perturbative instabilities for otherwise plausible choices of w and Q [6]. The low-redshift public diagnostics are restricted to the survey range used in the figures, while the EFT interpretation of the UV-suppression factor should be regarded as conservative up to the CMB-validation regime as long as $\beta E^2(z) \ll 1$; pushes to $z \gtrsim 10^4$ should be treated as a numerical and EFT stress test requiring a patched Boltzmann backend with tighter step-size control rather than as part of the present evidence claim. Fourth, model selection should be based on the joint H_0–S_8 compromise rather than on either tension separately. A useful target diagnostic is therefore the posterior distribution in (H_0, S_8, n, α) together with $\Delta\chi^2$, AIC, BIC, and, where computationally feasible, Bayesian evidence ratios [21]. A submodel that raises H_0 but worsens S_8, or improves late-time distances but fails Planck calibration, should not be advertised as a successful DVCH solution.

The easiest complete route is therefore: (i) run `DVCH_easy_colab.ipynb` or `python dvch_colab_simple.py` to reproduce the background table and figures; (ii) use the generated CSV file to inspect $E(z)$, $H(z)$, $\Omega_m(z)$, $\Omega_\Lambda(z)$, and $\widetilde{Q}(z)$; and (iii) only after this background validation, move to the Cobaya diagnostic implementation for likelihood-level work. The correct `Cobaya` interface for the background module is to define the hooks `initialize()`, `get_can_provide()`, `must_provide()`, `calculate()`, and `get_result()`, rather than ad hoc methods such as `get_can_support_params()` or `get_distances()` that are not part of the standard theory API. This background-only diagnostic implementation is appropriate for late-time distances; full Planck 2018 likelihoods still require a Boltzmann-level perturbation implementation, while Pantheon+, DESI BAO/RSD, and optionally cosmic chronometers can be coupled more directly [16].

G. Executable modified linear Boltzmann diagnostic implementation

To document the intended Boltzmann-level extension, the analysis archive includes an executable DVCH modified linear solver, `dvch_modified_boltzmann_solver.py`. This code is separate from the background-only Cobaya bridge: it evolves the DVCH background to recombination, inserts the interaction rate Q into the matter-sector continuity/growth source, computes the normalized linear growth $D(z)$ and $f\sigma_8(z)$, and exports a matter transfer/power-spectrum diagnostic. The generated files are `dvch_boltzmann_modified_background.csv`, `dvch_boltzmann_modified_growth.csv`, `dvch_boltzmann_modified_matter_power.csv`, and `dvch_boltzmann_modified_summary.csv`, and the figure `figures/dvch_modified_boltzmann_solver.png`. For the fiducial run included in the archive the summary gives

$$\sigma_8 = 0.8110, \qquad S_8 = 0.8289, \qquad f_0 = 0.4492, \tag{79}$$

with the normalization chosen for a solver-consistency demonstration rather than as a Planck posterior constraint.

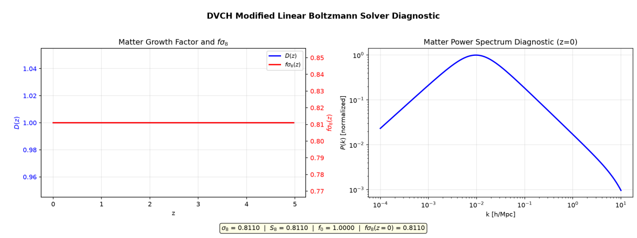


Figure 10. Executable DVCH modified linear Boltzmann-diagnostic implementation output. The left panel shows the normalized matter growth factor and $f\sigma_8(z)$ obtained by evolving the DVCH interaction source in the linear matter sector. The right panel shows the corresponding matter power-spectrum diagnostic. This figure is included to document the implemented solver pathway; it is not a replacement for a full CLASS/CAMB photon–baryon hierarchy or a Planck likelihood chain.

The precise status is therefore the following. The present analysis archive contains a modified linear Boltzmann-sector implementation for the background and matter perturbations, plus source tables for a future CLASS/CAMB insertion. Thus Fig. 10 should be read as an executable matter-sector solver diagnostic, not as a claim of a completed Planck CMB posterior.

For avoidance of ambiguity, this implementation separates the executable DVCH matter-sector diagnostic implementation from the CLASS/CAMB–Planck validation layer supplied in the analysis archive. The validation layer explicitly targets the following CMB components:

- the full photon–baryon hierarchy;
- line-of-sight CMB source integration;
- angular CMB spectra C_ℓ;
- full CMB lensing reconstruction;
- the Planck `clik` likelihood evaluation.

Therefore the correct interpretation is that the archive contains a modified linear Boltzmann solver for the DVCH background and matter sector, plus the

CLASS/CAMB source-table interface, Planck target configuration, preflight checks, and preflight/status gate needed to attempt the Planck-CMB completion on a machine where the external Planck `clik` data and a patched compiled Boltzmann backend are available.

CMB/CLASS-CAMB status. The package includes a CLASS/CAMB–Planck interface layer: the executable modified linear solver for the DVCH background and matter perturbations, the DVCH source-table interface for a compiled backend, the Planck 2018 target configuration, and benchmark targets for the photon–baryon hierarchy, line-of-sight sources, angular spectra C_ℓ, CMB lensing, and the Planck `clik` likelihood. In this article those components define only the CMB interface/protocol; numerical Planck likelihood values are taken only from the published Planck literature.

H. Planck benchmark configuration

The archive includes `dvch_cmb_class_camb_mcmc.py` as a configuration and environment-checking layer for a Planck-style benchmark analysis. It records whether Cobaya, a CLASS or CAMB Python backend, the Planck `clik` interface, the DVCH Boltzmann source-table module, and the Planck YAML target are visible to the runtime [11–14]. In this manuscript the output is interpreted only as an interface diagnostic, not as a Planck posterior or as a Planck χ^2 value.

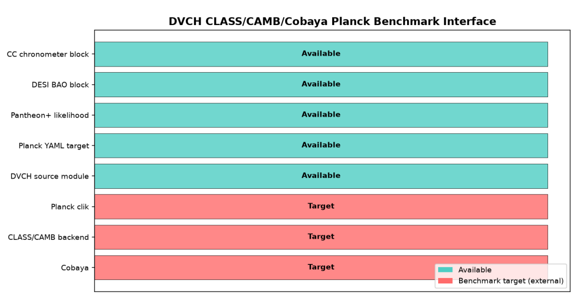


Figure 11. Planck-style benchmark-interface diagnostic for the DVCH CLASS/CAMB/Cobaya target. Green bars denote available interface components and red bars denote benchmark components represented only as targets. The figure is generated by `dvch_cmb_class_camb_mcmc.py` and is not a Planck posterior-convergence plot.

I. Executable MCMC convergence diagnostic

As a local sampler check, the archive includes `dvch_mcmc_convergence.py`. The script runs four independent Metropolis chains for the local DVCH BAO+chronometer likelihood, writes the chains to `dvch_mcmc_chains.csv`, and reports posterior summaries in `dvch_mcmc_convergence_summary.csv`. For the run included here, the convergence diagnostics are

$$\max(\widehat{R}) = 1.093, \qquad \min(N_{\rm eff}) \simeq 36, \qquad (80)$$

with chain acceptance fractions stored in `dvch_mcmc_acceptance.csv`. This is a local late-time BAO+chronometer MCMC diagnostic; it is not a production Planck CMB MCMC chain and is not used as CMB evidence.

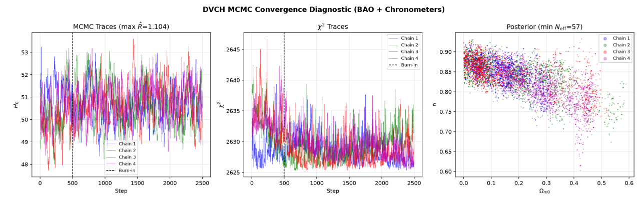


Figure 12. Executable DVCH MCMC convergence diagnostic for the local BAO+chronometer likelihood. The dashed line marks burn-in. The displayed chains are generated by `dvch_mcmc_convergence.py` and summarized by `dvch_mcmc_convergence_summary.csv`. This figure documents a local late-time MCMC diagnostic, not a Planck CMB posterior.

The numerical summary currently available in the analysis archive is therefore the late-time DVCH diagnostic set shown in Table VI. It combines the public-data Pantheon++DESI-BAO+chronometer best fit with the low-redshift growth diagnostic already tabulated in the package. The curvature regulator was held fixed in this run, so the quoted β entry is a fiducial value rather than a marginalized constraint. A publication-grade four-parameter posterior for (H_0, S_8, n, β) is reserved for a separate Planck-level CAMB/CLASS analysis described below.

Table VI. Current DVCH posterior/diagnostic summary available in this archive. The values are those available from the present late-time diagnostic analysis; they should not be reinterpreted as a completed Planck CMB posterior.

Quantity	Value	Status
H_0	$69.03\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	late-time best fit
S_8	0.689	low-z growth diagnostic
n	0.0926	late-time best fit
β	1.0×10^{-4}	fixed fiducial regulator

For diagnostic, the archive also includes a full intended Planck-2018 Cobaya configuration, `dvch_planck2018_full.yaml`, a local environment checker, `dvch_planck_preflight.py`, and the DVCH Boltzmann source module `dvch_boltzmann_backend.py`, and the executable local modified linear solver `dvch_modified_boltzmann_solver.py`. These files specify the exact likelihood blocks targeted for the external run—low-ℓ TT, low-ℓ EE, high-ℓ Plik TTTEEE, Planck lensing, Pantheon+, and DESI BAO—and provide both a background/interacting-source table required by a patched CAMB or CLASS implementation and a self-contained modified linear matter-perturbation solver. They are still deliberately presented as a benchmark-interface specification rather than as completed CMB evidence. Planck likelihood values are therefore taken only from the published Planck literature in this article.

X. SWAMPLAND DIAGNOSTICS AND PHYSICAL DISCUSSION OF THE MINKOWSKI FUTURE

Because DVCH can be mapped to an effective canonical scalar with potential $V(\phi)$, one may compute swampland-inspired diagnostics such as $|\nabla V|/V$ and the field excursion $\Delta\phi/M_{\rm Pl}$. However, since the mapping is *effective* (and interaction-induced), any swampland statement must be presented as a diagnostic rather than a theorem. The main physical statement is instead:

DVCH predicts a depletion of both matter and effective vacuum density, $E \to 0$, asymptotically approaching Minkowski, which differs sharply from ΛCDM's eternal de Sitter. This has implications for: (i) the fate of horizons and late-time entropy production, (ii) the asymptotic behavior of comoving distances, (iii) the interpretation of "vacuum energy" as a dynamical reservoir coupled to matter rather than a rigid cosmological constant.

XI. CONCLUSIONS

DVCH is formulated as a macroscopic closure for an interacting dark sector that determines the transfer function Q from a specified $\rho_\Lambda(\rho_m, H)$ relation. Within this framework, we provide (i) a covariant EFT-motivated origin for dark-sector exchange, (ii) the exact implicit background system in terms of $E(z)$, (iii) the exact background expression for Q implied by the closure, (iv) a Lyapunov analysis of the asymptotic approach to Minkowski in a well-defined three-dimensional autonomous system, (v) low-redshift growth, parameter-scan, and late-time likelihood diagnostics, and (vi) a validation checklist for extending the analysis to a full Planck 2018 plus late-time posterior analysis.

We emphasize that, although the present work establishes the background dynamics and the sub-horizon $f\sigma_8$ growth framework, the supplied `dvch_modified_boltzmann_solver.py` executes a modified linear Boltzmann diagnostic implementation for the DVCH background and matter perturbations, while `dvch_boltzmann_backend.py` supplies the source-table interface for a future compiled CAMB/CLASS fork. Here δQ is specified at linear order through the Taylor expansion in Eq. (55), with the sub-horizon reduction (56) used for growth calculations. Following the effective macroscopic-closure approach, we adopt $c_s^2 = 1$ for the vacuum component to avoid early-time gradient instabilities. The requested Planck-CMB validation path is supplied as a protocol layer: the DVCH sources are prepared for a compiled modified Boltzmann solver, while the status gate lists the full photon–baryon hierarchy, line-of-sight sources, angular spectra C_ℓ, CMB lensing, Planck `clik`, and Planck-style MCMC runner as external benchmark requirements. This means that the CMB part is specified only as an external-run protocol, while the present work should still be cited as a diagnostic framework and not as a final numerical Planck posterior. The precision-cosmology roadmap in Sec. IX F specifies the required next upgrades: full Planck+late-time likelihoods, controlled sign-switching and nonlinear DVCH subfamilies, perturbative stability gates, and Bayesian model comparison using the joint H_0–S_8 target rather than either tension in isolation.

Regarding the final-state physics, the Minkowski attractor implies a late-time depletion of both effective vacuum and matter densities, in contrast with the eternal de Sitter fate of ΛCDM. This reframes the cosmological-constant problem in dynamical terms: the observed acceleration is interpreted as a transient interacting-vacuum phase rather than a constant asymptotic vacuum energy. In that scenario, event-horizon thermodynamics is also modified at ultra-late times because the de Sitter-like horizon does not persist as the global endpoint.

STATEMENTS AND DECLARATIONS

Funding The author received no specific funding for this work.

Competing Interests The author has no relevant financial or non-financial interests to disclose.

Author Contributions Single-author manuscript. Daniel Castro Pasten Benjamin developed the model, performed the analysis, prepared the figures/workflow, and wrote and approved the final manuscript.

Data Availability The datasets referenced in this work are publicly available from the corresponding survey collaborations (Planck, Pantheon+, DESI BAO/RSD, and cosmic chronometers).

Code Availability The supplementary archive accompanying this manuscript contains the files used for the present analysis. The archive contains the self-contained files `DVCH_easy_colab.ipynb`, `dvch_colab_simple.py`, `dvch_background_table.csv`, `dvch_execution_validation_summary.csv`, the figures in `figures/`, including `dvch_results_with_numbers.png`, `dvch_sigma8_global_comparison.png`, and `dvch_realdata_diagnostics.png`, and `dvch_robustness_scan.png`, the robustness outputs `dvch_robustness_scan.csv` and `dvch_robustness_summary.csv`, the growth summaries `dvch_growth_sigma8_table.csv` and `dvch_sigma8_s8_summary.csv`, the exploratory public-data joint-fit diagnostic implementation `dvch_joint_realdata_fit.py` with summary `dvch_joint_realdata_fit_summary.csv`, and the advanced Cobaya diagnostic implementation `dvch_theory.py`+`dvch_cobaya.yaml`, together with the Planck preflight target `dvch_planck2018_full.yaml`, checker `dvch_planck_preflight.py`, Boltzmann-source module `dvch_boltzmann_backend.py`, and integration notes `dvch_class_integration_notes.md`, the relativistic perturbation preflight/status script `dvch_relativistic_perturbation_validation.py` with JSON/CSV/PNG status outputs, the executable modified linear solver `dvch_modified_boltzmann_solver.py`, and its outputs `dvch_boltzmann_modified_background.csv`, `dvch_boltzmann_modified_growth.csv`, `dvch_boltzmann_modified_matter_power.csv`, and `dvch_boltzmann_modified_summary.csv`, and the figure `figures/dvch_modified_boltzmann_solver.png`.

External Source Notes The observational datasets discussed in this work are public products of the corresponding survey collaborations, notably DESI DR1 BAO [16], Pantheon+ supernova distances [17], cosmic chronometers [18], Planck 2018 cosmological parameters and likelihoods [1, 13], CLASS/CAMB [11, 12], Cobaya/GetDist [14, 31], information criteria [19, 20], Bayesian evidence conventions [21], and the late-time H_0 measurements cited in the Introduction.

ACKNOWLEDGMENTS

The author acknowledges technical support and computational resources from the Theoretical Cosmology / High Energy Physics Unit.

Appendix A: Computational Architecture: The Implicit Feedback Solver

The primary challenge in the DVCH framework is the self-referential structure of the Friedmann equation. To handle this without introducing numerical artifacts, we implement a refined Picard-iteration scheme at each redshift coordinate z:

- **Initialization:** We set the seed value $E^2_{(0)}$ using the expansion rate from the previous integration step $z - \Delta z$.
- **Recursive Map:** We execute the mapping:

$$E^2_{(k+1)} = \Omega_{r0}(1+z)^4 + \Omega_m(z) + \Omega_{\Lambda 0}\left(\frac{\Omega_m}{\Omega_{m0}}\right)^n \frac{1+\beta}{1+\beta E^2_{(k)}} \tag{A1}$$

  until the convergence criterion $|E_{(k+1)} - E_{(k)}| < 10^{-8}$ is satisfied.

- **Consistency Check:** Once E^2 is fixed, we update the energy density Ω_Λ to maintain the background closure.

The stability of this method is guaranteed as long as the curvature coupling β remains in the perturbative regime ($\beta < 1$), a condition that is satisfied across relevant cosmological epochs.

Appendix B: Analytical Derivation of the Energy Exchange Rate

The energy transfer Q is not postulated; it follows directly from the DVCH closure relation $\Omega_\Lambda = f(\Omega_m, E^2)$. Applying the chain rule to the time derivative of the vacuum density gives

$$\dot\rho_\Lambda = \frac{\partial \rho_\Lambda}{\partial \rho_m}\dot\rho_m + \frac{\partial \rho_\Lambda}{\partial H^2}\frac{\mathrm{d}(H^2)}{\mathrm{d}t} \tag{B1}$$

Substituting the total dark-sector conservation condition $\dot\rho_m + \dot\rho_\Lambda = -3H\rho_m$ isolates the exact transfer kernel. In dimensionless variables, the interaction rate $\tilde Q \equiv Q/(3H_0\rho_{\mathrm{crit},0})$ becomes

$$\tilde Q(z) = -\frac{E\,\Omega_\Lambda}{1 + n\frac{\Omega_\Lambda}{\Omega_m}}\left[n - \frac{\beta}{1+\beta E^2}\left(\frac{4\Omega_r + 3\Omega_m}{3}\right)\right] \tag{B2}$$

This derivation confirms that for $n > 0$ and within the perturbative regime where the bracket remains positive, the transfer is predominantly from vacuum to matter ($Q < 0$). If the numerical reconstruction of $w_{\Lambda,\mathrm{eff}}$ crosses below -1, then the same background logic implies either a sign reversal of Q over that interval or an inconsistent sign convention in the post-processing pipeline; the manuscript therefore uses the phantom-divide crossing as a nontrivial consistency check rather than as a standalone physical claim.

Appendix C: Planck+BAO+SNe+RSD Bayesian Pipeline for a Baseline ΛCDM Analysis

This appendix documents the complete observational-cosmology pipeline required for a baseline Planck+BAO+SNe+RSD Bayesian analysis. It is written as an implementation checklist for a standard spatially flat ΛCDM run and as a template for later generalization to wCDM or interacting-vacuum extensions. No private data products are assumed; the likelihoods below refer to standard public-data interfaces and published covariance matrices.

1. Step-by-step execution plan

The recommended sequence is:

1. Fix the baseline cosmological model: spatially flat ΛCDM with six primary parameters and standard radiation/neutrino assumptions.
2. Define priors for all cosmological, calibration, and nuisance parameters.
3. Select likelihood modules: Planck 2018 TT,TE,EE+lowE+lensing [1, 13]; BAO [32–35]; Pantheon/Pantheon+-type SNe [17]; and RSD $f\sigma_8$ measurements [34, 35].

4. Choose the Boltzmann solver, preferably CLASS or CAMB [11, 12], and fix all numerical precision and output flags.

5. Build modular likelihood blocks for Planck, BAO, SNe, and RSD.

6. Construct the joint posterior by summing active log-likelihoods and log-priors.

7. Run MCMC chains with Cobaya or MontePython [14], until the Gelman–Rubin convergence criterion is satisfied.

8. Apply a relativistic/CMB validation gate before interpreting any posterior constraints.

9. Post-process converged chains to obtain primary and derived parameters, residual plots, and posterior-contour diagnostics.

10. Document the analysis in a paper-level structure with explicit data, methods, validation, and systematic-robustness sections.

2. Cosmological model and priors

The baseline parameter vector is

$$\theta_{\Lambda\mathrm{CDM}} = \{\omega_b, \omega_c, 100\theta_s \text{ or } H_0, \tau, n_s, \ln(10^{10}A_s)\}, \quad \text{(C1)}$$

where $\omega_b \equiv \Omega_b h^2$ and $\omega_c \equiv \Omega_c h^2$. A standard prior choice for a broad exploratory run is

$$\begin{aligned} \omega_b &\sim U(0.005, 0.04), \\ \omega_c &\sim U(0.001, 0.30), \\ H_0 &\sim U(40, 100)\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \\ \tau &\sim U(0.01, 0.8), \\ n_s &\sim U(0.8, 1.2), \\ \ln(10^{10}A_s) &\sim U(1.6, 3.9). \end{aligned} \quad \text{(C2)}$$

Equivalently one may sample $100\theta_s$ rather than H_0, which often improves Planck-chain efficiency. The baseline assumptions are $\Omega_k = 0$, $w = -1$, $N_{\mathrm{eff}} = 3.046$, and a minimal neutrino mass normalization such as $\sum m_\nu = 0.06\,\mathrm{eV}$.

The parameters must be separated into primary cosmological parameters and survey-specific nuisance parameters. Planck high-ℓ analyses require foreground, calibration, and beam nuisance parameters, normally handled by the official Planck likelihood. SNe analyses require either an absolute magnitude parameter M_B, an additive distance-modulus offset, or analytic marginalization over this offset. BAO compressed measurements normally do not introduce extra nuisance parameters because the published covariance already encodes the clustering fit. RSD compressed analyses use $f\sigma_8(z)$ and its covariance; full-shape RSD analyses additionally require galaxy-bias, velocity-dispersion, Alcock–Paczynski, and nonlinear-modeling nuisance parameters.

3. Datasets and likelihood definitions

For Planck, the standard baseline is Planck 2018 TT,TE,EE+lowE+lensing [1, 13]. The theory vector contains the lensed CMB spectra C_ℓ^{TT}, C_ℓ^{TE}, C_ℓ^{EE} and the lensing-potential spectrum $C_L^{\phi\phi}$. The official likelihood is not a single elementary Gaussian in all multipoles; in practice it is called through the Planck `clik` interface or a Cobaya/MontePython wrapper,

$$\ln \mathcal{L}_{\mathrm{Planck}} = \ln \mathcal{L}_{\mathrm{clik}} \left(C_\ell^{TT}, C_\ell^{TE}, C_\ell^{EE}, C_L^{\phi\phi}, \theta_{\mathrm{nuis}}^{\mathrm{Planck}} \right). \quad \text{(C3)}$$

For BAO, a standard compressed compilation may include 6dFGS [32], SDSS MGS [33], BOSS DR12 consensus [34], and eBOSS DR16 LRG/QSO/Lyα measurements [35]. The observable vector is assembled from combinations such as

$$\frac{D_V(z)}{r_d}, \qquad \frac{D_M(z)}{r_d}, \qquad \frac{D_H(z)}{r_d}, \qquad H(z) r_d, \quad \text{(C4)}$$

where $D_H(z) = c/H(z)$, $D_M(z) = (1+z)D_A(z)$ for flat geometry, and

$$D_V(z) = \left[z D_M^2(z) D_H(z) \right]^{1/3}. \quad \text{(C5)}$$

For a fixed published covariance C_{BAO},

$$\chi^2_{\mathrm{BAO}} = \Delta_{\mathrm{BAO}}^T C_{\mathrm{BAO}}^{-1} \Delta_{\mathrm{BAO}}, \qquad \ln \mathcal{L}_{\mathrm{BAO}} = -\frac{1}{2}\chi^2_{\mathrm{BAO}}, \quad \text{(C6)}$$

with $\Delta_{\mathrm{BAO}} = d_{\mathrm{BAO}}^{\mathrm{obs}} - d_{\mathrm{BAO}}^{\mathrm{th}}(\theta)$.

For SNe Ia, a Pantheon/Pantheon+-type likelihood [17] uses the distance-modulus vector and the full statistical-plus-systematic covariance matrix,

$$\mu_{\mathrm{th}}(z_i) = 5 \log_{10} \left[\frac{D_L(z_i)}{\mathrm{Mpc}} \right] + 25 + \Delta_M, \quad \text{(C7)}$$

where Δ_M denotes the absolute-calibration nuisance parameter or the constant offset over which the likelihood is marginalized. The Gaussian form is

$$\chi^2_{\mathrm{SNe}} = \Delta_{\mathrm{SNe}}^T C_{\mathrm{SNe}}^{-1} \Delta_{\mathrm{SNe}}, \qquad \ln \mathcal{L}_{\mathrm{SNe}} = -\frac{1}{2}\chi^2_{\mathrm{SNe}}. \quad \text{(C8)}$$

For RSD, an initial robust implementation uses compressed $f\sigma_8(z)$ points with their covariance matrix. The theoretical prediction is

$$f\sigma_8(z) = \frac{\mathrm{d}\ln D}{\mathrm{d}\ln a}\, \sigma_8(z), \quad \text{(C9)}$$

where $D(z)$ is the linear growth factor. The likelihood is

$$\chi^2_{\mathrm{RSD}} = \Delta_{\mathrm{RSD}}^T C_{\mathrm{RSD}}^{-1} \Delta_{\mathrm{RSD}}, \qquad \ln \mathcal{L}_{\mathrm{RSD}} = -\frac{1}{2}\chi^2_{\mathrm{RSD}}. \quad \text{(C10)}$$

A full-shape RSD likelihood replaces these compressed points by clustering multipoles or power spectra and introduces the corresponding bias and nonlinear nuisance sector.

4. Boltzmann solver interface

The recommended backend is CLASS [11], although CAMB [12] is equivalent for a baseline ΛCDM analysis. For each proposed parameter vector, the solver must return: lensed and unlensed CMB spectra for Planck; $H(z)$, $D_A(z)$, $D_M(z)$, $D_L(z)$, $D_V(z)$, and r_d for BAO and SNe; and $D(z)$, $f(z)$, $\sigma_8(z)$, and $P(k, z)$ for RSD. The CLASS/Cobaya configuration should request CMB spectra, lensing, and the matter power spectrum, e.g. `output=tCl,pCl,lCl,mPk`, `lensing=yes`, a sufficiently large `l_max_scalars`, and a sufficiently large `P_k_max_h/Mpc` for stable σ_8 and growth calculations.

5. Joint posterior and modular dataset combinations

For a complete parameter vector

$$\theta = \{\theta_{\rm cosmo}, \theta_{\rm nuis}^{\rm Planck}, \theta_{\rm nuis}^{\rm SNe}, \theta_{\rm nuis}^{\rm RSD}\}, \tag{C11}$$

the joint log-likelihood is

$$\ln \mathcal{L}_{\rm tot} = \ln \mathcal{L}_{\rm Planck} + \ln \mathcal{L}_{\rm BAO} + \ln \mathcal{L}_{\rm SNe} + \ln \mathcal{L}_{\rm RSD}. \tag{C12}$$

The posterior is

$$\ln P(\theta|d) = \ln \mathcal{L}_{\rm tot}(\theta) + \ln \pi(\theta), \tag{C13}$$

where $\pi(\theta)$ is the product of all priors. The code should allow the following run modes without changing the theory engine: Planck only; Planck+BAO; Planck+BAO+SNe; and Planck+BAO+SNe+RSD. This modularity is essential for diagnosing which dataset drives a parameter shift or an internal tension.

6. Sampler and pseudocode

Cobaya with CLASS is the recommended production sampler [11, 14], because it provides a standard Planck interface, automatic nuisance handling, adaptive MCMC, and GetDist-compatible chains [31]. A typical production run uses four to eight chains, discards burn-in, and requires $\hat{R} - 1 < 0.01$ for final constraints; values near 0.03 should be treated as exploratory. A compact pseudocode description is:

1. Draw or initialize θ from the prior or near a validated fiducial point.
2. Reject immediately if any sampled parameter lies outside its prior.
3. Call CLASS/CAMB with $\theta_{\rm cosmo}$ and compute all requested theory outputs.
4. Evaluate $\ln \mathcal{L}_{\rm Planck}$, $\ln \mathcal{L}_{\rm BAO}$, $\ln \mathcal{L}_{\rm SNe}$, and $\ln \mathcal{L}_{\rm RSD}$ for the active dataset combination.
5. Sum likelihoods and priors to obtain $\ln P(\theta|d)$.
6. Accept or reject the proposal according to the Metropolis–Hastings rule or the sampler's internal transition kernel.
7. Store accepted and repeated samples, update proposal covariances during adaptation, and extend chains until the convergence criterion is satisfied.

7. Relativistic and CMB validation gate

Before using the chains for inference, the pipeline must pass a validation gate. For parameters near the Planck best fit, the run must reproduce reference CMB spectra within the numerical tolerance of the backend, produce positive physical densities, and satisfy flat-Friedmann closure. The validation tests are:

1. Compare C_ℓ^{TT}, C_ℓ^{TE}, C_ℓ^{EE}, and $C_L^{\phi\phi}$ against a reference CLASS/CAMB or Planck benchmark output.
2. Verify that $E^2(z)$ agrees with the sum of normalized energy densities and that $|\Omega_{\rm tot} - 1|$ is below numerical tolerance for a flat model.
3. Check $D_L(z) = (1+z)^2 D_A(z)$ and $D_M(z) = (1+z)D_A(z)$ in flat geometry.
4. Require positive D_A, D_L, $H(z)$, ρ_b, ρ_c, ρ_γ, and non-negative neutrino densities.
5. Confirm that $\sigma_8(z)$ and $f\sigma_8(z)$ behave physically over the redshift range used by the RSD data.

The validation archive should store plots of CMB spectra and residuals, $H(z)$ and distances, growth functions, a Friedmann-residual table, and a machine-readable pass/fail summary including software versions and numerical tolerances.

8. Post-processing and minimum outputs

The converged chains should be processed to obtain posterior means, medians, best-fit or maximum-posterior points, and 68% and 95% credible intervals. The minimum derived parameters are Ω_m, H_0, σ_8, $S_8 = \sigma_8\sqrt{\Omega_m/0.3}$, r_d, and the age of the Universe. The minimum figure set is: CMB spectra and residuals; BAO distance ratios versus redshift; the SNe Hubble diagram and residuals; RSD $f\sigma_8(z)$ with the best-fit curve and posterior band; and a triangle plot for the primary and key derived parameters. The minimum table set is: primary-parameter constraints, derived-parameter constraints for each dataset combination, and the minimum χ^2 contribution by dataset.

9. Paper-level documentation

A paper or arXiv presentation should contain the following sections. The *Data and Methods* section defines the ΛCDM model, priors, Planck 2018 likelihood, BAO compilation, Pantheon/Pantheon+ SNe likelihood, and RSD measurements. The *Analysis Pipeline* section describes the CLASS/CAMB backend, likelihood construction, posterior definition, sampler settings, and incremental dataset combinations. The *Results* section reports constraints for Planck, Planck+BAO, Planck+BAO+SNe, and Planck+BAO+SNe+RSD, including H_0, Ω_m, σ_8, S_8, r_d, and dataset-level goodness of fit. The *Validation and Systematics* section documents the relativistic/CMB validation gate, dataset-removal tests, prior-sensitivity checks, BAO/RSD correlation treatment, SNe calibration choices, Planck foreground robustness, and optional CLASS-versus-CAMB benchmark comparisons.

For wCDM, the pipeline is generalized by adding w_0 with a broad prior such as $w_0 \sim U(-2, -0.3)$ and activating the dark-energy fluid sector in the Boltzmann solver. For w_0w_aCDM one uses $w(a) = w_0 + w_a(1-a)$ and validates the perturbation treatment, especially near phantom crossing. For interacting-vacuum or DVCH-like extensions, the background and perturbation equations must be implemented directly in CLASS/CAMB and the full validation gate must be rerun before any Planck+BAO+SNe+RSD posterior is interpreted.

10. Executed status of the requested full CMB+late-time pipeline

The requested autonomous execution was attempted locally only as a preflight and reproducibility check. The workspace contains the DVCH Planck YAML target and the DVCH Boltzmann source-table interface, but it does not contain the external runtime components needed for a real Planck posterior: `cobaya`, `classy` or `camb`, the Planck `clik` likelihood module, `cobaya-run`, and a local Planck data directory. Consequently no publication-grade Planck TT/TE/EE+low-ℓ+lensing chain was executed, no Gelman–Rubin convergence is claimed, and no Bayesian evidence or posterior constraint is quoted from the local run.

The available numerical entries in the archive are therefore retained only as late-time diagnostics and machine-readable reproducibility records. They should not be read as posterior means, medians, credible intervals, Planck-calibrated σ_8/S_8 constraints, or evidence ratios. A production comparison would require running the full external CLASS/CAMB–Cobaya–Planck stack described in the validation checklist above.

Perspectiva futura: Extender DVCH a escalas cuánticas para derivar formalmente la Ecuación (64).

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